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Leveraging Large Language Models for Automated Web Scraping: A Focus on Academic Career Planning

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Abstract

In the highly competitive academic landscape, where the number of applicants far exceeds the available professorial positions, planning an academic career path is a complex task. Crucial data on relevant job positions is scattered across numerous platforms and job portals, in addition to individual job listings on university homepages. This vast and ever-changing data landscape must be laboriously searched and compiled manually, which often leads to errors and incomplete results. Furthermore, tracking potential future vacancies presents an additional challenge. Therefore, a solution is needed that can automatically collect, organize, and present this information efficiently and user-friendly, thereby potentially increasing academics' chances of finding a professorship that matches their skills and aspirations.

This thesis aims at designing and implementing an innovative, automated system for collecting, filtering, and presenting data about academic job positions that can aid academics in career planning. The proposed solution needs to leverage both traditional web scraping approaches and state-of-the-art Large Language Models (LLMs) to extract and consolidate publicly accessible information. The data should be pulled from both structured and unstructured data sources, only utilizing publicly accessible web resources such as DFG GERiT, DBPedia, search engines, and university homepages, with the scope set to academic opportunities within the D-A-CH region (Germany, Austria, Switzerland). The necessary data points for each position include, but are not limited to: Country, State, University, Type, Faculty/Department, Institute, Chair, Professorship, URLs, Job Holder, Appointment Date, Grouping (W1-W3), Vacancy, and Application Deadlines.

The objective of this thesis is the creation of a solution or the combination of existing techniques to solve the problem of automatically providing academic career planning information based on methods from the fields of Web Scraping, Data Science and Natural Language Processing (NLP) as described above. This comprises the analysis of the state of the art of suitable extraction methods as well as the demonstration of the solution by implementation and an appropriate experimental evaluation assessing the accuracy and completeness of the data extraction, and the usability of the result representation.

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List of Abbreviations

API	Application Programming Interface
BPMN	Business Process Model and Notation
D-A-CH	Germany, Austria, Switzerland
DFG	Deutsche Forschungsgemeinschaft (German Research Institutions)
FOSS	Free and Open-Source Software
HR	Human Resources
LLM	Large Language Model
MIT	Massachusetts Institute of Technology
N/A	Not Applicable
NLP	Natural Language Processing
OR	OpenRouter
PhD	Doctor of Philosophy
SUS	System Usability Scale
UEQ	User Experience Questionnaire
WCAG	Web Content Accessibility Guidelines

1 Introduction

In the dense and highly competitive landscape of academic job search, especially in the D-A-CH region (comprising Germany, Austria and Switzerland), it has become increasingly difficult to obtain a professorship. Although Germany is the world's third largest producer of scientific knowledge and publishes more articles than almost any other country except the U.S. and China, the employment situation for academics remains complex [1]. It is alarming that Germany has one of the lowest numbers of academics in permanent employment in all of Europe. The job market for academics with doctorates is even described as a "bottleneck". Those who fail to make the leap to a professorship often find themselves "caught between two stools", in a state of professional limbo that offers little stability or fulfillment. [2]

In a broader context, 90% of researchers employed in German academia work on fixed-term contracts, many of which last less than a year. This trend affects young scientists, including PhD students, postdocs, and senior researchers seeking permanent professorships. The current academic environment, characterized by hyper-competition, short-term prospects, and uncertain contract renewals, creates an environment that often leads to an outflow of highly competent and motivated young scientists. [3]

This scenario highlights the urgent need for innovative solutions to optimize the job search process for academics, especially those who have earned their PhD and are seeking a professorship. Against this background, this thesis aims to make an important contribution to solving this problem.

1.1 Current Situation

Planning for an academic career, especially towards a professorship, involves a multitude of information from various sources. This includes data about current vacancies, potential future positions, appointment dates, and more. However, this information is often scattered across multiple platforms and presented in various formats. These platforms range from job portals such as Academics.de, ResearchGate Jobs, and StepStone,

to individual job portals or listings on university homepages, among others. The heterogeneity of data representation on these platforms further complicates data collection.

The complexity of this search is worsened by the increasingly competitive nature of academic careers. According to a report by the Federal Statistical Office *(Statistisches Bundesamt)*, over 28,000 students earned their PhD in Germany in 2021 alone [4]. This rush of qualified candidates contrasts with a relatively stagnant number of available professorial positions. According to the GENESIS database, there were only about 50,000 professorial positions in total in the same year, a number that is growing only slightly each year [5]. To make matters worse, many of these professorial positions are often filled for long periods of time, sometimes for several decades or until they reach retirement age. These long-term tenures further limit the already limited opportunities for new entrants to academia, exacerbating the challenges of finding an academic job.

This imbalance in the number of PhD graduates relative to available professor positions is not unique to Germany. In Canada, for instance, the number of PhD graduates has nearly doubled from 2002 to 2017, yet the number of tenured professor positions remained stable since 2009 with 41,000 positions [6]. Moreover, these findings also extend to the United Kingdom. A report published by the Royal Society in 2010 revealed that only 0.45% of PhD graduates in the UK transition into a professorship. The vast majority of these graduates, transition to roles outside of academia at various stages of their academic careers [7]. These international trends highlight the scarcity of professorial positions relative to the number of aspirants.

When searching for a job, academics today typically turn to spreadsheet programs such as Excel to manually enter and track a variety of career-related data. This method involves compiling information from a variety of platforms - from dedicated academic job portals to individual university homepages. While Excel provides a familiar environment for organizing data, it is far from optimized for this particular type of information aggregation. But as simple as this manual approach may seem, it comes with limitations that raise questions about its sustainability and efficiency.

1.2 Motivation

The current manual process for collecting and organizing job-related information in academia is subject to several significant problems. First and foremost, the labor intensity of the process cannot be overstated. Academics spend a great deal of time manually

searching for, extracting, and compiling relevant data-a process that becomes more laborious the longer the planning typical of academic careers.

In addition to being time-consuming, the manual nature of the process makes it prone to human error. Errors in data entry or misinterpretation of information can lead to missed opportunities or misdirected efforts. In addition, the problem extends to the completeness and accuracy of the data itself. The academic work environment is in constant flux, with potential future jobs as important as current ones. This requires constant, careful monitoring of numerous sources to track current positions and predict upcoming vacancies-a task that is not only tedious, but also prone to oversights and inaccuracies.

From an aesthetic perspective, spreadsheet tools like Excel are not designed for visual appeal or ease of interpretation in the context of complex, multivariable data. This means that academics often struggle to present the data they collect in a way that is intuitively understandable and actionable. Hence, there is a need for an automated solution that can facilitate this burden, gathering, organizing, and presenting the required information in an efficient and user-friendly manner. Such a solution could streamline the process, potentially increasing the chances of PhDs securing professorships that are an ideal fit for their skills and aspirations.

1.3 Problem Analysis

The central problem is to automate the information search for PhD job positions with the focus on professorial positions. This task presents multiple challenges due to the sheer variability and dynamic nature of the data sources and the constant need for up-to-date information.

A main hurdle lies in the heterogeneity and distribution of data sources. These sources range across job portals, search engines, and university websites, each presenting its own format and set of attributes. Currently, there is no single comprehensive repository that brings together all of the positions relevant to professorships in a country or even in a specific academic area. This scattered layout is further complicated by the unique structures of different academic institutions, as faculties, chairs, and professorships can vary widely in terms of how they are organized and labeled.

This complexity is amplified by the dynamic character of the data. Listings about academic positions is anything but static; it is constantly updated and needs to be reviewed

regularly. Manual attempts to keep track of this ever-changing landscape are not only labor intensive, but also prone to errors and omissions, compromising the overall reliability and completeness of the data.

At many institutions, professorships are often held over an extended period of time, sometimes several decades, until the incumbent professor eventually retires. This longevity of tenure creates an environment in which opportunities for advancement within the academic hierarchy are significantly limited. Therefore, prospective academics could greatly benefit from being able to predict when a professorial position will become available in order to plan and strategize their careers. However, forecasting these vacancies poses a significant challenge.

One main obstacle in predicting vacancies is the limited availability of structured, publicly available data that could provide reliable estimates. For example, information on the age of a current professor, a critical variable for predicting the likelihood that a position will become vacant due to retirement, is often not readily available to the public. This information gap complicates efforts to anticipate and prepare for potential career opportunities in academia.

The final point to consider is the usability of the data. The accumulation of large amounts of data is only useful if it is presented in a user-friendly format that allows effective searching, filtering, and interpretation. Here the challenge lies not only in managing the representative aspects of the data, but also in creating mechanisms for meta-analysis. For example, tools for analyzing trends in job placements or the career paths of existing professors could provide invaluable insights for academics' career planning.

Overall, these issues underscore the necessity for an automated solution capable of navigating and efficiently utilizing this complex data landscape for the benefit of PHDs.

1.4 Objectives and Scope

The goal of this thesis is to develop an innovative, automated system for collecting, filtering, and presenting data about academic job listings to simplify the complicated process faced by academic professionals. This system envisions a seamless integration of conventional web scraping techniques with the sophisticated capabilities of Large Language Models (LLMs). Such integration aims to ensure accurate extraction and consolidation of relevant information from structured and unstructured public web resources,

focusing on platforms such as DFG GERiT[1], DBPedia[2], job portals, search engines, and individual university homepages.

To measure the efficiency and reliability of the proposed solution, the extracted data is benchmarked against a carefully constructed manually created ground truth. Beyond data precision, a major focus is on usability. Given the multi-layered needs of the academic community, the final product should not only provide accurate data, but also be insightful, contextual, and intuitive - a goal further supported with a detailed questionnaire presented to a representative sample group.

The geographic scope of this work is on the D-A-CH region. This region, home to a large number of universities, consists of 423 universities in Germany [8], 160 in Austria [9], and 42 in Switzerland [10], providing a significant sample size for this research. Expected data points to be extracted include country, state, university, faculty/department, institute, chair, professorship, URLs, job holder, appointment date, grouping (W1 - W3), job offer, and application deadline. Methodologically, the research will draw on areas such as web scraping, data science, and natural language processing (NLP).

Although this thesis aims to provide a comprehensive solution, certain limitations must be acknowledged. The focus on the D-A-CH region naturally excludes academic labor markets outside Germany, Austria and Switzerland. Moreover, no meta-analyses are conducted on trends in academic job vacancies or on the mobility patterns of professors between different institutions. Beyond the technological intricacies and methodologies, this research essentially represents an effort to bridge the gap between the academic community and the positions they seek, and to ensure that professionals are equipped with comprehensive, reliable, and user-oriented tools in the highly competitive academic landscape of the D-A-CH region.

1.5 Outline

In the first part of this thesis, we introduce the topic of career planning for academics, providing a brief overview of the associated challenges and setting out the objectives of this research. We then transition into the State-of-the-Art chapter, a deep dive into current approaches and best practices. We define key requirements for academic career

[1] https://www.gerit.org/de
[2] https://www.dbpedia.org/

planning and analyze existing solutions against these requirements to identify challenges.

The next chapter presents a conceptual approach outlining a proposed solution to the challenges identified. This includes addressing potential limitations that might arise during implementation. Following this, the implementation chapter discusses the practical realization of our solution based on the conceptual approach. We detail the technologies and strategies adopted to build a solution that supports academic career planning especially for postdocs.

The evaluation chapter (see 5) objectively assesses the performance of our solution. Our solution undergoes thorough testing against a carefully curated ground-truth, as well as a real-world user study that captures user feedback on the usability of our solution.

In the last chapter we conclude our work and findings and its limitations as well as possible solutions to these limitations and potential areas of future research.

2 State-of-the-Art

This chapter aims to describe the requirements necessary for an optimal solution in the field of academic career planning, drawing insights from the objectives and scope of this thesis outlined in the previous chapter. Considering these problems and requirements, the following sections provide an analysis of established methods and strategies that are widely used in this field. The goal of this investigation is to evaluate the effectiveness and relevance of these existing approaches in the context of the requirements set.

This assessment will serve as the basis for identifying gaps and potential areas of improvement in the area of academic career planning, while also providing us with a framework for comparing our implemented solution in a later phase of this work.

2.1 Requirements Analysis

An effective solution must meet six criteria to best support professors, postdocs, and senior PhD students in navigating the complexities of the academic job market. For this master's thesis, we have identified a set of 6 requirements (see Table 2.1), against which existing groups of solutions will be evaluated in relation to the problem statement posed in 1.3.

In the following chapters, we will elaborate on each of these requirements. In addition, we introduce an assessment scale with the following ratings: N/A (Not Applicable), ★☆☆☆ (Insufficient), ★★☆☆ (Poor), ★★★☆ (Fair), and ★★★★ (Good). The N/A rating is included to recognize situations where specific criteria do not apply to a particular solution. This scale is used to systematically evaluate solutions based on their compliance with these criteria.

Requirement	**Requirement**
R1 Low-Effort	R4 Data Completeness
R2 Ease-of-Use	R5 Data Refresh and User Preference Retention
R3 Versatile Data Integration	R6 PhD-Centric Solution

Table 2.1: Catalog of Requirements

2.1.1 R1: Low-Effort

This requirement is intended to ensure that the system minimizes user input and maximizes output efficiently and effectively. In academia, where time is a limited resource, this criterion is particularly important. Professors, postdocs, researchers, and faculty members are often overwhelmed with numerous tasks - from teaching to research to administrative duties [11]. In such a time-pressured environment, a system that requires a lot of manual input and technical expertise can be a bottleneck rather than a relief.

Therefore, the main goal of this requirement is to ensure that the system should be as autonomous as possible. This is achieved by reducing the need for manual data entry, avoiding complicated customization steps, and streamlining complex operations. Users should be able to focus on interpreting and using the data collected, rather than dealing with data entry and navigating the system.

The evaluation of state-of-the-art methods with respect to this requirement is performed using the following assessment scale (see Table 2.2).

Rating	Condition
★☆☆☆	The solution relies entirely on manual data entry, for example into spreadsheets, which requires a lot of work.
★★☆☆	The solution requires technical expertise to configure, and it requires regular user input for searches, filters, or data updates.
★★★☆	While the solution does not require technical expertise, it still requires regular user input for searches, filters, or data updates.
★★★★	After initial configuration, the solution fully automates data collection and updating without the need for recurring human effort.

Table 2.2: Assessment Scale for R1

2.1.2 R2: Ease-of-Use

The Ease-of-Use Requirement (R2) aims to create a user-centric system that minimizes barriers to use. In the academic world, where users are often multi-tasking and have varying levels of technical expertise, a user-friendly system is not just a convenience, but a necessity. This requirement identifies six key criteria to ensure that the system is intuitive, accessible, and efficient to use for all users:

- **Search functionality:** An important success criterion that evaluates the system's ability to provide simple and effective search options without requiring technical syntax or expertise. [12]
- **Accessibility:** Supported by the Web Content Accessibility Guidelines (WCAG), this criterion ensures that the system meets common accessibility standards so that for example people with disabilities can work effectively with the system. [13]
- **Export functionality:** The focus of this criterion is whether it is possible to export an overview or an Excel spreadsheet with potential job listings.
- **Documentation:** The presence of a help section, FAQs, or documentation that can help users navigate the system effectively is evaluated here. [14]
- **Recommendations:** This assesses whether the system offers suggestions or recommendations to help users in finding potential job listings.
- **Multi-Device Support:** This criterion evaluates the system's ability to work across mobile and desktop devices.

The evaluation of state-of-the-art methods with respect to this requirement is performed using the following assessment scale (see Table 2.3).

Rating	Condition
★☆☆☆	The system meets **0 to 1 out** of 6 criteria.
★★☆☆	The system meets **2 to 3 out** of 6 criteria.
★★★☆	The system meets **4 to 5 out** of 6 criteria.
★★★★	The system meets **6 out of 6** criteria.

Table 2.3: Assessment Scale for R2

2.1.3 R3: Versatile Data Integration

This requirement underscores the need for the system to use only data from publicly available and verifiable platforms to ensure compliance with ethical standards, privacy laws or institutional policies. Examples of such platforms include DFG GERiT, DBPedia, and university homepages, among other publicly available resources.

The use of publicly available data not only guarantees compliance with legal and ethical standards, but also ensures the verifiability and reliability of the information used by the system. These approaches provide a transparent and comprehensive basis for the research and data, increasing the utility and reliability of the system.

Furthermore, adopting public data not only improves the reproducibility of the solution by ensuring that results and methods can be independently validated and tested, but also ensures the sustainability and adaptability of the system. Future work and subsequent projects can easily adopt and build enhanced solutions on top of the available, public data, thereby promoting the continuous improvement and optimization cycle.

The evaluation of state-of-the-art methods with respect to this requirement is performed using the assessment scale in Table 2.4.

Rating	Condition
★☆☆☆	Less than 25% are publicly accessible and verifiable.
★★☆☆	Between 25% and 50% are publicly accessible and verifiable.
★★★☆	Between 51% and 75% are publicly accessible and verifiable.
★★★★	More than 75% are publicly accessible and verifiable.

Table 2.4: Assessment Scale for R3

2.1.4 R4: Completeness of the Data

This requirement focuses on the ability of the system to provide a comprehensive set of data points for each academic position. It is important to provide not only basic details, but also a complete and differentiated overview of the available options. Comprehensive job listings should cover the following data points:

- **Country:** The nation where the academic position is located.
- **State/Region:** Specific state or region within the country.
- **University:** The name of the educational institution offering the job.
- **Type of Institution:** The type of institution offering the academic position, such as a university or a university of applied sciences.
- **Faculty/Department:** The faculty or department within the university.
- **Institute:** The specific institute or research center within the university.
- **Chair:** If the professorship or the professor holds a chair *(Lehrstuhlinhaber)*.
- **Professorship:** The type or name of the offered professorship.
- **URLs:** Web links for further information or application.
- **Job Holder:** Current or last known holder of the position, if applicable.
- **Appointment Date:** When the position starts, if applicable.

- **Grouping (W1 - W3):** The salary grouping (W1-W3) of the position.
- **Vacancy Status:** Whether the position is open or occupied.
- **Vacancy Date:** The expected or estimated date of vacancy.
- **Application Deadline:** Last date for submitting applications.

In the highly competitive academic job market, complete information gives job seekers an advantage in designing their applications and preparing for interviews. This also eliminates the need for users to consult multiple data sources, thus saving valuable time in the job search process.

The evaluation of state-of-the-art methods with respect to this requirement is performed using the assessment scale in Table 2.5.

Rating	Condition
★☆☆☆	**0 Items**: No data points are available.
★★☆☆	**1-5 Items**: Only a limited number of data points are available.
★★★☆	**6-10 Items**: Moderate range of data points are available.
★★★★	**11-15 Items**: Nearly all essential data points are covered.

Table 2.5: Assessment Scale for R4

2.1.5 R5: Data Refresh and User Preference Retention

The purpose of this requirement is to ensure that the solution provides an optimal user experience by integrating two essential features: dynamic data updating and personalized user preferences. Fulfilling these two aspects in an excellent way will ensure that the system not only keeps its information relevant and up to date, but also provides a customized experience that meets the individual needs and expectations of the users.

Job listings are dynamic data that are subject to constant change in terms of availability, application deadlines, and requirements. The effectiveness of a system in this context depends heavily on its ability to update this data in real time or at regular intervals. Failure to do so can result in outdated listings remaining on the platform, causing users to miss opportunities or waste time searching for jobs that are no longer available. Timely data updates are therefore not just a feature, but a critical requirement for maintaining user trust and engagement.

As data is routinely updated to ensure current and relevant content, there is a risk of losing personalized settings or marked favorites. For academic users features that allow them to mark items as favorites are of great importance. Whether these marks identify coveted research positions, desired universities, or a preferred professorship, maintaining them permanently during data refresh cycles is critical.

Meeting this requirement is critical to the overall success and adoption of a system. A platform that fails in updating data or retaining user preferences is likely to suffer from lower user activity and potentially high bounce rates. A system that is superior in both aspects is therefore more likely to attract and retain a solid user base, thereby solidifying its market position.

The evaluation of state-of-the-art methods with respect to this requirement is performed using the assessment scale in Table 2.6.

Rating	Condition
★☆☆☆	The solution neither allows data updates nor retains user preferences.
★★☆☆	The solution retains user preferences but does not permit data updates.
★★★☆	The solution allows for data updates but fails to retain user preferences.
★★★★	The solution automatically updates the data and retains user preferences.

Table 2.6: Assessment Scale for R5

2.1.6 R6: PhD-Centric Solution

The purpose of this requirement is to ensure that the proposed system is not just another job search tool, but a specialized platform tailored to the specifics of the academic job market faced by postdocs and senior PhD students. As they transition from doctoral to a professorship, these individuals face a unique set of challenges and requirements that differ from those of the broader job seeker population. Unlike general job search platforms that often cover a wide range of roles or are designed for HR departments, this system is intended to serve as a targeted resource for PhDs.

One of the most distinct challenges in the academic job market is its extended timeframe. From the time a position is advertised to the time it is filled, the process can span months. In addition, PhDs are often juggling multiple job offers at once, so they need to keep track of the various application stages, deadlines, and requirements over an extended

period. A system that can help with this long-term tracking, perhaps through robust alerting and filtering capabilities, would be invaluable to this user group.

However, the complexity does not end with the duration of the job search. The academic job market is not a monolith; it encompasses a range of positions, from research roles to professorships. Doctoral students often pay particular attention to potential professorial vacancies, as these are associated with both prestige and tenure. A system that can prioritize or highlight such positions, or better yet, offer a search tailored to the user's academic specialization, will provide a significant advantage in the job search.

The evaluation of state-of-the-art methods with respect to this requirement is performed using the assessment scale in Table 2.7.

Rating	Condition
★☆☆☆	Solution does not target the search for job postings, making it unsuitable for PhDs searching a professorship.
★★☆☆	Solution targets job search but does not focus on the academic community, thereby missing the unique needs of PhDs.
★★★☆	Solution targets job search within the academic community but is not specifically designed to address the unique needs and challenges faced by PhDs.
★★★★	Solution is specifically targeted to address the unique job search needs of PhDs, including specialized information, long-term opportunity tracking, and a focus on academic vacancies.

Table 2.7: Assessment Scale for R6

2.2 Approaches

The following section aims to evaluate different approaches for the academic job search based on the requirements defined in the previous section. These approaches can be divided into different categories, each of which has several characteristics, advantages, and limitations. Specifically, we focus on these six main groups show in Table 2.8.

Each of these groups will be evaluated according to the requirements outlined earlier, ranging from low effort and ease of use to the specific needs of doctoral students. These ratings will be based on our previously introduced rating scale (see 2.1) and the agreement with the particular requirement. The results of this analysis are central to this

thesis. They will be used in the design and implementation of a solution specifically tailored to the unique job search needs of professors, postdocs, and senior PhD students.

Approach	Description
Job Portals	Websites or platforms that specialize in listing job openings.
University Homepages	Official websites of academic institutions listing current job openings.
NLP-based	Tools that use Natural Language Processing to identify job openings.
Web Scraping	Automated scripts to collect job listings from various websites.
Search Engine	Platforms like Google and Bing for broad and targeted job searches.
API	Provide structured access to job listings through a common interface.

Table 2.8: Approach Groups

2.2.1 Job Portals

Job portals serve as a central platform where employers can post job ads and job seekers can search these ads to find suitable positions. These platforms are often equipped with various features such as advanced search filters, job alerts and company profiles. While job portals are versatile and cover a wide range of industries and functions, there are also those that specialize in academic and research-related positions, that we will focus on here. Some popular examples in the academic context include portals like Academics.de, Euraxess, Jobs-beim-Staat.de, ResearchGate Jobs, and ScholarshipDB.

Academics.de, operated by the ZEIT publishing house, is a German job portal specialized in jobs for academic, science and research positions. It offers a broad range of job opportunities, from research and postdoc positions to professorships across a wide range of universities. The platform enables users to utilize keywords in their job search and filtering options for sector, position, and work area, among others (see Figure 2.1). In addition, the platform offers valuable resources such as career tips and articles tailored to academics. While the primary focus is on the German-speaking academic community, Academics.com offers an English equivalent, even though most of the job ads and descriptions appear to be mostly in German. [15]

Figure 2.1: Overview of the Academics.de Job Portal

Euraxess is a European initiative that provides a comprehensive platform for researchers, entrepreneurs, and research organizations. In addition to thousands of job listings for a wide range of fields and qualifications, ranging from research assistants over doctorial positions to professorships, it also provides information on funding opportunities, networking events, relocation services and general career advice. The job search functionality also offers features as searching by keywords and filtering options for the research area, researcher profile and country, among others. Its broad European focus is highlighted by its multilingual support, which offers content in 24 languages to appeal to academics who seek opportunities across the EU. [16]

Jobs-beim-Staat.de focuses on public sector job opportunities in Germany, including academic positions at government-funded institutions. While the primary focus of this platform is Germany, it also includes job listing for other German speaking countries. Users can navigate through the job listings with simple search and filter options, making it easier to find relevant offers. The portal's clear focus on the public sector, including academia, making it a central platform for specialized listings, although it is mainly operated in German. [17]

ResearchGate Jobs is an extension of the ResearchGate network, a professional network for academics and researchers. The Jobs section features listings that are often posted directly by researchers and academic institutions, making it a reliable source for specialized academic jobs. In addition, the platform offers the opportunity to connect with potential employers and colleagues in the respective research areas, and it is primarily available in English, although with some multilingual job postings.

ScholarshipDB is unique in that it offers not only job opportunities, but also a wide range of scholarships, fellowships, and research grants. This makes it a versatile platform for academics at different stages of their careers, from students seeking scholarships to professionals looking for research funding. [18]

LinkedIn and Xing have evolved into robust job search functions in addition to their role as professional networking platforms. These business- and employment-focused social media platforms not only create a dynamic environment where job postings are interwoven with professional networking features, but also enable precise search and filters that help users narrow down opportunities relevant to their academic and professional careers. By supporting multiple languages, both appeal to a broad, international audience and ensure accessibility and usability for diverse demographics. [19]

2.2.2 University Homepages

University homepages serve as the official online presence of academic institutions and often include a dedicated section for job openings at the university or affiliated research centers. These platforms provide a direct channel for job seekers by delivering first-hand information directly from the source. Job postings on university homepages typically cover a wide range of academic roles, from faculty positions to research assistant positions.

Due to their directness and specificity, the homepages of universities differ significantly from general job portals. The job offers are exclusive to the academic institution itself and are often neither aggregated nor listed on alternative platforms. Listings on university homepages are usually very detailed and provide job seekers with all the important information they need to fully understand the position, requirements, and associated deadlines, however they are often offered in an unstructured format tending to differ on a university-to-university basis.

In addition, these listings are usually exceptionally current, as they are updated directly by the institution, ensuring that potential applicants are viewing the most recent and relevant job openings. Also, university homepages are usually provided in multiple languages, at least German and English versions are available for most university homepages in the D-A-CH area.

2.2.3 NLP-based

With the rapid progress of the digital age, data extraction and analysis methods have also evolved. Natural Language Processing (NLP) is at the forefront of these advances, providing sophisticated tools that enable the transformation of voluminous and unstructured textual data into meaningful insights. In the context of job search, NLP-based approaches represent an innovative solution from traditional methods by leveraging computational power to automatically analyze, understand, and categorize job postings from multiple sources. While robust and innovative, these approaches span a spectrum from ready-to-deploy applications to experimental solutions, with each approach bringing its own opportunities and challenges.

NLP has been successfully incorporated into various applications across the recruitment and job search domain, albeit with varying focus and user-centricity:

- **Recruiter-Centric Applications:** Platforms like HireVue[3], Metaview[4], Tengai Unbiased[5], and PeopleScout[6] prioritize the recruiter's view, enhancing interview analytics and candidate assessment, thereby enriching human resource (HR) departments' productivity and bias mitigation.
- **Job Seeker-Centric Applications:** Tools like Talentprise[7] and Kickresume[8] employ NLP to process user profiles or CVs, assisting in pinpointing matching job opportunities. Notably, LoopCV[9] automates the process even further, not only identifying but also applying to relevant jobs based on the user's CV, bridging the gap between passive search and active application.

However, it is important to note that despite their advanced capabilities LLMs are not flawless. They can sometimes create so-called "hallucinations" - information that seems plausible but is false or misleading. This is a major problem in the context of job applications. For example, if an LLM is used to analyze job listing, there is a risk that it will create inaccurate or misleading information. This can lead to inappropriate job matches or overlook potential opportunities, impacting both recruiters and job seekers. [20], [21]

[3] https://www.hirevue.com/
[4] https://www.metaview.ai/
[5] https://tengai.io/
[6] https://www.peoplescout.com/
[7] https://www.talentprise.com/
[8] https://www.kickresume.com/en/
[9] https://www.loopcv.pro/

The continued evolution of NLP has also spawned experimental job search applications that combine web scraping with NLP models to refine job search. "Job Search NLP", for example, combines a web scraper that collects data from Indeed with an NLP model that enables semantic search for job ads [22] Similarly, an approach developed by Jim King et al. involves developing a web app that reads job ads from Indeed and then uses NLP to process and generate summaries that provide an overview of the job market for specific tech positions in well-known tech cities in the US. [23]

While these experimental approaches illustrate the potential of integrating NLP into automated job search processes, they also highlight some challenges. These solutions often require a certain level of technical expertise for the institution and are predominantly tailored to platforms such as Indeed, which limits their applicability especially for job listings in the D-A-CH area. In addition, these tools are not tailored to the specific needs of niche groups, such as postdocs. While NLP-based approaches represent a promising way to automate and improve job search processes, their current applications cannot provide a one-size-fits-all solution, especially for specialized academic job searches.

2.2.4 Web Scraping

Web scraping-based solutions employ automated scripts that extract job postings from various online platforms, including academic websites, job boards, and even social media. Web scraping offers a more proactive method of capturing job postings because it pulls data directly from multiple sources to centralize it in one place. Given the vast number of job listings scattered across the internet, web scraping can be an effective way to ensure that no opportunity is overlooked.

One of the main benefits of web scraping is its ability to aggregate different types of data. Whether it is a specialized academic job portal or a general career website, web scraping tools can efficiently gather and compile this information. However, this method often requires a certain level of technical expertise, as scraping tools may need to be configured to target specific data fields such as job title, location, and academic discipline.

Some notable examples in the field of web scraping for job postings include Octoparse, Zyte, Propellum, and Apify, among others:
Octoparse is characterized by a user-friendly interface and a versatile range of applications. Octoparse can read data from various websites without requiring programming knowledge, making it accessible to a large number of users. Noteworthy is its ability to

prevent or bypass rate limiting and anti-scraping techniques, while extracting data from dynamic websites through JavaScript rendering. Besides, it allows downloading data in various traditional formats, such as CSV, JSON and Excel. [24]

Zyte, formerly known as Scrapinghub, is another platform that offers a range of web scraping tools, including a cloud-based platform for running scrapers. It is geared more towards professionals and enterprises and offers additional services such as data extraction and crawling. Zyte not only offers to run custom scrapers, but also allows users to purchase data directly by asserting that they "probably already extract job postings data from your target job boards". This implies an extensive database from various job boards is ready for use. In terms of data transfer, Zyte offers formats such as CSV, JSON, JSON-Lines and XML. [25]

Propellum offers specialized job scraping services tailored to the needs of job boards and recruitment agencies. The platform not only collects job ads, but also enriches the data with additional details and insights. It aims to provide an end-to-end solution for job data aggregation, enrichment, and distribution. Moreover, they provide API access and supports common data delivery formats like CSV, JSON, HTML, and XML. [26]

Apify is a platform that offers a wide range of web scraping and automation solutions, including a marketplace for pre-built scrapers. These scrapers can be used to collect job postings via an API that is specific to the job board being scraped. Since most scrapers are created by individual developers, there is no standard output format. Some of the scrapers only support the API, while others can also export data in JSON, XML, CSV, Excel, and HTML formats. The platform is designed for both individual developers and larger enterprises and offers a range of customized solutions. [27]

2.2.5 Search Engines

Unlike web scraping, search engine-based approaches use the extensive capabilities of search engines such as Google and Bing to locate job postings. Such platforms are incredibly flexible and provide current data by indexing a wide range of publicly accessible online content, including website texts and metadata. While search engines share some similarities with web scrapers in terms of aggregating information, they expand beyond mere data collection to process, rank, and synthesize various data points. This results in a more nuanced search experience.

An important aspect of these approaches is their advanced search function, which allows users to execute tailored queries. This capability proves especially beneficial for job searches since it enables both general and specific inquiries. Incorporating specific keywords such as "Professorship" can narrow search results to academic job listings. Additionally, the flexibility of search engines allows for catering to a range of search needs beyond the job market.

Google, a leading search engine, has improved job searches through its specialized feature, Google Jobs[10]. This feature consolidates job postings from numerous online platforms, streamlining the job search process. Although Google Jobs is helpful for fast searches, it may lack the specific filters and capabilities offered by academic career portals. Nevertheless, its expansive coverage and user-friendly interface make it a valuable tool for job seekers.

It's worth noting that the search results are primarily tailored for human readability and user experience, rather than automated processing. Search engines such as Google and Bing prioritize information presentation in a format that users can easily understand and navigate. However, this user-centric design implies that although the information is accessible and easy to comprehend, it lacks features for automated data management or collection. These platforms do not include features like bookmarking favorite results or tools for data extraction and analysis. While this user-friendly design is advantageous for job seekers, it may present difficulties for those seeking to automate data processing or integrate the data with other software systems.

2.2.6 APIs

API-based approaches offer a more direct and structured method for accessing job data, distinguishing themselves from web scraping methods. Leveraging the capabilities of Application Programming Interfaces (APIs) provided by different job boards and career platforms, these approaches facilitate streamlined and efficient access to job listings.

One noteworthy example in this field is the **Arbeitnow.com Job Board API**. The API is distinctive in that it consolidates information from numerous job board APIs into a single, unified interface. This aggregation makes it a powerful tool for accessing a wide range of job listings from multiple sources through a single API endpoint. [28]

[10] https://www.google.com/about/careers/applications/

Greenhouse.io is a major player in the job board industry with its Job Board API, allowing direct access to a broad range of job listings. This API is valuable for organizations that want to integrate job listings into their own websites or applications, providing real-time updates on job openings. [29]

SmartRecruiters.com offers an API that focuses on aiding employers in managing and posting job listings across multiple platforms. This enables a centralized recruiting approach that expands the job post's scope. [30]

The **LinkedIn Job Posting API** also plays a major role, especially for employers and recruiters. as it enables efficient management of job postings on LinkedIn, capitalizing on its vast network of professional users. [31]

The APIs provided by Arbeitnow.com, Greenhouse.io, SmartRecruiters.com, and LinkedIn primarily cater to general job seekers and do not specialize in niche areas, such as the academic job market or job offerings for professorships.

2.3 Comparison

After an in-depth examination of various job search approaches - from traditional platforms such as job portals and university homepages to more technologically advanced methods such as web scraping and NLP-based solutions - we now come to one central part of this state-of-the-art: the comparison. The goal of this section is to contrast these different approaches considering the requirements we have formulated, and to provide a comprehensive analysis that highlights their respective strengths and weaknesses, particularly in the context of job search for postdocs.

We will evaluate the extent to which each group of approaches meets the previously stated requirements. By comparing these groups side-by-side, we aim to provide a multi-dimensional perspective that will be helpful in developing and implementing a job search solution that truly meets the needs of postdocs.

2.3.1 R1: Low-Effort

In the following section, we evaluate and rate each approach based on its alignment with the Low-Effort requirement (see 2.1.1).

Job Portals Rating: ★★★★ (Good)
Job portals are designed with user-centricity and automation at their core. After initial setup, e.g. configuring search parameters such as "professorship" and geographic preferences such as "Germany", these platforms can automatically aggregate new listings and notify users via email or other notification mechanisms. The intuitive user interface and minimal technical knowledge required to set up searches play a key role in making job portals efficient and effective, and minimizing the time and effort required of the user after initial setup, resulting in a “good” rating.

University Homepages Rating: ★☆☆☆ (Insufficient)
Navigating through university homepages requires a significant manual effort, as these platforms generally do not offer automation features to streamline the job search process. Users must manually search, compile, and regularly update job postings, which is very time-consuming. In addition, this method requires a constant effort to keep up to date with new opportunities, which is not in line with the low effort requirement, resulting in a “insufficient” rating.

NLP-Based Approaches Rating: ★★☆☆ (Poor)
While NLP-based approaches emphasize automation, particularly in terms of reducing the workload of HR departments, they are inadequate in the context of the low effort requirement due to the technical expertise required to set up and maintain them. Although they can automate certain tasks, they require manual updating of job data and regular training of models to maintain accuracy and relevance, which requires constant user intervention and a certain level of technical expertise, resulting in a “poor” rating.

Web Scraping Rating: ★★☆☆ (Poor)
Web scraping approaches offer a degree of automation, as they can be configured to collect and update job posting information on their own. However, initial setup is not trivial and requires expertise in configuring scraping parameters and ensuring ethical and legal compliance. While this eliminates the need for constant manual data entry, the technical hurdle to setting up and maintaining the system is very high, resulting in a “poor” rating.

Search Engines Rating: ★★★☆ (Fair)
Search engines, while not specifically designed for job searching, offer a simple and user-friendly interface that requires minimal effort on the part of the user. By inputting pertinent keywords, individuals can peruse an extensive range of job postings from diverse sources. This procedure necessitates no specialized technical expertise and is highly

accessible. However, the absence of precise job search characteristics and the requirement to manually filter through possibly irrelevant results means that while search engines are easy to use for initial searches, they may require more effort to find precisely relevant job listings, resulting in a "fair" rating.

APIs Rating: ★★☆☆ (Poor)
APIs offer a potent approach for obtaining and merging job listings from a range of sources. Nevertheless, they lack user-friendliness for non-professionals due to the considerable technical expertise and programming skills required for deployment and upkeep. APIs provide vast customization opportunities and the possibility of automation, but their complex nature and the need for ongoing maintenance make them a poor fit for the low-effort requirement. This tool is better suited for technical users or organizations with IT resources rather than for individuals seeking user-friendly job search tools, resulting in a "poor" rating.

2.3.2 R2: Ease-of-Use

In the following section, we evaluate and rate each approach based on its alignment with the Ease-of-Use requirement (see 2.1.2).

Job Portals Rating: ★★★★ (Good)

- **Search functionality:** Job portals generally offer a robust and user-friendly search functionality that does not require technical syntax.
- **Accessibility**: Due to their user-centric design and adherence to web standards, they generally ensure accessibility, including support possibly screen readers.
- **Export functionality:** The lack of export functionality is a notable shortcoming.
- **Documentation:** Comprehensive help texts or FAQs are usually available.
- **Recommendations:** They often contain job or relevant suggestions.
- **Multi-device support:** They usually offer seamless use across desktop and mobile devices.

Despite the lack of export functionality, the job portals meet most of the criteria, so the rating of "good" is justified.

University Homepages Rating: ★★☆☆ (Poor)

- **Search functionality:** Often lack specific job search functionality.
- **Accessibility:** Generally, they comply with web standards and ensure accessibility.

- **Export functionality:** The lack of a dedicated export function is common.
- **Documentation:** A FAQ section is usually present, although not always comprehensive.
- **Recommendations:** They usually do not offer job recommendations.
- **Multi-device support:** They generally offer cross-device support.

The universities' homepages meet some criteria but have significant weaknesses in key areas such as search functions and recommendations, which justifies a "poor" rating.

NPL-Based Rating: ★★☆☆ (Poor)

- **Search functionality:** Mixed performance; clear in applications focused on users but complicated in others.
- **Accessibility:** Some solutions centered on users often align with accessibility standards, while others may not.
- **Export functionality:** Available in some solutions but not in all.
- **Documentation:** Often lacking in experimental solutions, available in recruiter-focused tools.
- **Recommendations:** Present in job seeker-focused applications, absent in others.
- **Multi-device support:** User-focused solutions offer this, while technical ones may not.

Overall, NLP-based solutions show potential but have notable limitations in areas like search functionality, recommendations, and consistent user experience, leading to a "poor" rating.

Web Scraping Rating: ★☆☆☆ (Insufficient)

- **Search functionality:** Web scraping does not inherently provide search functionality.
- **Accessibility:** Since they are not user-oriented tools, they often lack accessibility features.
- **Export functionality:** They allow data export in various formats and thus meet this criterion.
- **Documentation:** A documentation is often available but may require technical knowledge to fully understand.
- **Recommendations**: No recommendations are provided.
- **Multi-device support**: Usually do not support use across multiple devices.

Web Scraping meets only a few criteria and is clearly inadequate in terms of intuitive and user-friendly operation, which is why it is rated "insufficient".

Search Engines Rating: ★★★☆ (Fair)

- **Search functionality:** Strong capabilities, but not tailored specifically for job searching.
- **Accessibility:** High level of accessibility through adherence to web standards and assistive technologies.
- **Export functionality:** No direct export capabilities for search results or job listings.
- **Documentation:** General, non-job search specific help and support documentation available.
- **Recommendations:** Provide personalized search results, not personalized job recommendations.
- **Multi-device support:** Optimized for mobile and desktop platforms.

Search engines provide a level of ease of use with broad accessibility and multi-device support; however, their job-specific features and recommendations are constrained, which is why it is rated as "fair".

APIs Rating: ★☆☆☆ (Insufficient)

- **Search functionality**: Customizable but technically demanding.
- **Accessibility:** Not designed for direct use of end-users, lacking in web accessibility standards.
- **Export functionality:** Flexible data retrieval and formatting options.
- **Documentation:** Comes with a technical documentation, not intended for end-users.
- **Recommendations:** Dependent on the application's use of API data, recommendations are typically not offered by the API.
- **Multi-device support:** APIs are accessible across multiple devices as they lack a user interface.

APIs are technically focused and not user-friendly for the average PhD job seeker, resulting in a "insufficient" rating, despite their sophisticated backend capabilities.

2.3.3 R3: Versatile Data Integration

In the following section, we evaluate and rate each approach based on its alignment with the Versatile Data Integration requirement (see 2.1.3).

Job Portals Rating: N/A

Job portals are known for compiling offers from a variety of sources, often including both public and private data. Much of this private data comes from companies or institutions that manually create listings to find potential employees. While these portals offer a wide range of job listings, the lack of transparency in their sources makes it difficult to determine the origin of these listings, whether from private or public data sources.

Nevertheless, it is plausible to assume that a significant number of these listings come directly from publicly available platforms such as corporate websites, university homepages, or other public job portals. In addition, it is likely that the job portals rely on databases such as GERiT of the German Research Foundation (DFG) to provide a comprehensive overview of the structure of research institutions in Germany. However, due to the uncertainty of the origin of these data sources, we rate this approach as “N/A” (not applicable).

University Homepages Rating: ★★★★ (Good)
University homepages are the primary sources that provide publicly available and verifiable first-hand information about academic job openings. The nature of this information ensures its suitability and adaptability and provides a solid foundation for future efforts and projects. Based on these characteristics, we rate this approach as "good".

NLP-based Rating: N/A
NLP-based solutions show great diversity in terms of their data sources. While some rely predominantly on public data, especially those using web scraping techniques such as Job Search NLP, others rely on private databases or APIs. This bifurcation makes it difficult to ensure the verifiability and public accessibility of the full range of data. As with the job portal approaches, this approach is rated N/A (not applicable) because we cannot verify the data sources.

Web Scraping Rating: ★★★★ (Good)
Web scraping approaches primarily focus on publicly available data, especially when configured to extract information from platforms such as university homepages or job boards. The inherent flexibility of most web scraping solutions allows for manual customization and facilitates the extraction of data from multiple sources, including databases such as GERiT or DBPedia and university homepages. Given the transparent and publicly available nature of these data sources, we award a rating of "good" for this approach.

Search Engines Rating: ★★☆☆ (Poor)

Search engines provide access to a wide range of public and private data sources, from academic job listings on university websites to corporate career pages and online job boards. However, their lack of transparency and control over data sources presents a significant challenge. Users frequently encounter difficulties in determining the origin, authenticity, and accuracy of data. The ranking algorithms utilized by search engines sometimes give higher priority to less relevant or less current sources. Users frequently encounter difficulties in determining the origin, authenticity, and accuracy of data. Due to these limitations, particularly when it comes to verifying data sources, using the search engine approach is deemed "poor".

APIs Rating: ★★★★ (Good)
APIs provide a more dependable and regulated method for data integration in comparison to search engines. A number of APIs are supplied by reliable sources, such as public websites or job portals, which ensures a greater degree of data authenticity and reliability. The usage of APIs enables the retrieval of data from these sources in a direct and systematic manner. The challenge of utilizing APIs stems from their varying degrees of diversity, access, and data quality. While certain APIs may offer extensive and current data, others may be restricted in their scope or suffer from data lags. Additionally, reliance on third-party providers for APIs can impede data continuity and consistency. Despite these challenges, APIs generally offer a good level of data integration, especially when it comes to verifiable and authoritative sources. Thus, this approach is rated as "good".

2.3.4 R4: Completeness of the Data

In the following section, we evaluate and rate each approach based on its alignment with the Completeness of the Data requirement (see 2.1.4).

Job Portals Rating: ★★★★ (Good)
Job portals, which are known for their extensive offerings, usually provide a wide range of data, including the country of the job, the affiliated university, faculty or department details, URLs, and even application deadlines. Despite their broad scope, specific details such as the type of institution, the type of department, or possible deadlines for applications are sometimes missing. Such details may be included in the text descriptions or require another look at the university's official website. However, since most of the essential data points are covered, this approach deserves a "good" rating.

University Homepages Rating: ★★★★ (Good)
The homepages of the universities are the most important sources for academic job advertisements and usually offer comprehensive data. They contain not only precise information about the university, faculty, institute and chair, but also important data such as the application deadline. Some fine details, such as expected dates for filling positions, may not always be apparent, since job announcements are usually posted when a position is about to be filled or is newly opened. Given the detail and comprehensiveness of the data provided, this approach is rated "good".

NLP-based Rating: ★★★☆ (Fair)
NLP-based approaches, which analyze text from various sources to extract job data, vary significantly in their effectiveness. While they can identify a wide range of details, such as job roles, university information, and deadlines, their performance is highly dependent on the quality of the source material. These methods excel at extracting nuanced information but face challenges in maintaining consistency across diverse formats and detailedness of data sources. Due to these variable factors, NLP-based approaches receive a "fair" rating for completeness of data.

Web Scraping Rating: ★★★★ (Good)
Web scraping is inherently versatile and capable of extracting a large number of data points, especially when they come from detailed sources such as university homepages. However, the effectiveness of this method depends largely on the specific websites being scraped and the complexity of the scraping process. Because web scrapers can be used to gather data from a variety of sources, including job portals and university homepages, they offer both potential and challenges, resulting in a “good” rating.

Certain challenges remain, however. Ensuring consistency and accuracy across a variety of sources and offerings can be complicated given the different formats and representations on different websites. Nonetheless, web scraping approaches cover nearly all of the key data points mentioned in the requirement and provide job seekers with a detailed overview of academic positions. Due to its ability to provide a holistic overview, the web scraping approach is rated "good".

Search Engines Rating: ★★★☆ (Fair)
Search engines offer a gateway to a wide range of resources, such as job portals, university websites, and news articles pertaining to academic positions. Their robustness stems from their ability to collate data from a variety of sources. Nevertheless, the comprehensiveness of the data differs considerably based on the search query and the indexed

sources. Search engines often provide links to job postings, but they may not include important details, such as application deadlines, department specifics, or institutional information. Search engines are considered "fair" for completeness of data due to their broad but non-specialized nature.

APIs Rating: ★★★★ (Good)

APIs designed for academic job listings provide a structured approach to data retrieval. They are tailored to specific data sources, such as university databases and professional academic networks, allowing for efficient retrieval of detailed and relevant data. APIs offer various data points, including job location, university details, faculty information, and application deadlines, among others. However, the level of data completeness chiefly relies on the source from which the API acquires information. If the source is in-depth, the API can transmit a broad array of data. Difficulties may occur when the source data is restricted or when consolidating data from numerous APIs to form a comprehensive dataset. Nevertheless, APIs mostly provide a high degree of data completeness, resulting in a "good" rating in this category.

2.3.5 R5: Data Refresh and User Preference Retention

In the following section, we evaluate and rate each approach based on its alignment with the Data Refresh and User Preference Retention requirement (see 2.1.5).

Job Portals Rating: ★★★★ (Good)
Job portals are the epitome of dynamic platforms, designed from the beginning to update their data. They offer users the ability to set preferences, save searches, and mark preferred listings. Because of their design, these portals are able to store users' preferences and ensure that offers are up to date. This not only provides a tailored user experience, but also helps maintain user engagement. Based on these features, we give this approach a rating of "good".

University Homepages Rating: ★★★☆ (Fair)
University homepages, while providing direct and often up-to-date information, may fall short in terms of personalization features. They normally lack the ability to save user preferences, search queries, or preferred listings, which can detract from the overall user experience. Although a rich source of up-to-date job listings, especially for the academic community, the lack of personalization features leads us to rate this approach as "fair".

NLP-based Rating: ★★★★ (Good)
NLP-based methods by their nature can be designed to accommodate the dynamic nature of data. They have the potential to update data frequently and maintain user-oriented preferences, depending on their specific implementation. This dynamic and user-centric nature is often observed in NLP solutions tailored to HR and job seekers. Such methods not only keep the information up to date, but also address the personalized needs of the users. Given the combined benefits of dynamic updating and maintaining user preferences, NLP-based approaches are rated as "good".

Web Scraping Rating: ★★★☆ (Fair)
Web scraping is a powerful tool for data extraction, especially for obtaining fresh and updated job listings. These methods are designed to access data dynamically, ensuring that the job ads displayed are always up to date. Where web scraping weakens, however, is in the area of user personalization. Most web scraping solutions are primarily data-driven and do not have integrated user interfaces or platforms that enable a customized user experience. The lack of features that allow users to save searches or mark preferred listings can be a significant disadvantage for those seeking a personalized job search experience. Consequently, while web scraping approaches are excellent in terms of data timeliness, their limited ability to personalize users results in a "fair" score.

Search Engines Rating: ★★★★ (Good)
Search engines are good at providing up-to-date information. This is due to their sophisticated algorithms and huge index of web content. They continuously scan and update their databases, guaranteeing that users have access to the most recent data. Additionally, contemporary search engines have customization capabilities, enabling users to customize search results according to their preferences, search history, and even location. This feature enhances the user's experience by filtering job opportunities based on individual preferences. While some search engines excel in refreshing data, they may not effectively retain specific user preferences for job searches, which slightly lowers their rating. Overall, we rate search engines as "good" in fulfilling the requirements for data refresh and user preference retention.

APIs Rating: ★★★☆ (Fair)
APIs, particularly those designed for job listing and recruitment platforms, play a crucial role in providing updated data. They allow for real-time data retrieval from various sources, ensuring that the information is current. However, the degree of user preference retention largely depends on the application that utilizes these APIs. While APIs themselves are capable of delivering fresh data, they often lack direct user interaction

features, such as saving searches or preferences. This limitation is primarily due to their role as data conduits rather than user-facing platforms. As such, APIs score a "fair" rating, excelling in data refresh but falling short in direct user preference retention.

2.3.6 R6: PhD-Centric Solution

In the following section, we evaluate and rate each approach based on its alignment with the PhD-Centric Solution requirement (see 2.1.6).

Job Portals Rating: ★★☆☆ (Poor)
While job portals are extensive and diverse, they often lack specialized sections dedicated exclusively to the academic community, particularly doctoral students seeking professorships. While some platforms include filters or sections for academic jobs, they are generally not attuned to the particular dynamics and long deadlines of the academic job market. Although they are aimed at a wider audience, they have a limited capacity to meet the specific needs of postdocs, which justifies the rating of "poor".

University Homepages Rating: ★★★☆ (Fair)
University homepages, especially those of larger research institutions, are a goldmine of job postings for PhDs. They often offer direct, specific, and detailed academic job postings that include coveted faculty positions such as professorships. However, they do not offer advanced features such as long-term tracking or alert systems for future vacancies. Because of their focus on academic job postings, especially those relevant to PhDs, they receive a "fair" rating.

NLP-based Rating: ★★☆☆ (Poor)
The technological sophistication of NLP-based approaches is undeniable. Nevertheless, their focus on the academic community, and on the particular challenges faced by postdocs, is not a given. Unless explicitly developed with the academic job market in mind, these approaches provide only general job search functions that, while efficient, may not fully address the differentiated needs of postdocs. This leads us to rate them as "poor".

Web Scraping Rating: ★★☆☆ (Poor)
Web scraping is essentially a powerful tool for extracting academic job listings from a variety of platforms. It can be fine-tuned to locate jobs that are of interest to PhDs. However, because it is a standalone tool that does not provide a user-centric platform with specific features such as tracking or notification systems, it may not be ideal for meeting

the tedious and complex needs of doctoral researchers. Therefore, the rating of "poor" is appropriate.

Search Engines Rating: ★★☆☆ (Poor)
Search engines such as Google offer comprehensive job search capabilities, but they lack specialized features geared toward PhDs. These engines aggregate job postings from various sources, including academic positions. Nonetheless, they lack tools specific to the academic job market, like detailed insights or tailored alerts aimed at industry positions. Although the broad coverage is convenient, the absence of personalized support for the unique needs of doctoral students gives these search engines a rating of "poor".

APIs Rating: ★★★☆ (Fair)
APIs from job platforms and academic institutions enable the creation of custom job search solutions for PhDs. These applications can specifically target academic job postings, including features like deadline tracking and field-specific notifications. Their flexibility and potential for customization align closely with the nuanced needs of PhDs. However, their effectiveness relies heavily on available technical expertise for development and maintenance. APIs are considered "fair", reflecting their high potential for targeted support, albeit with a technical prerequisite.

2.4 Conclusion

In this chapter, we summarize the key findings and propose the most optimal solution for professors, postdocs, and senior PhD students as we have completed our investigation of different job search methods. Our analysis comprehensively compares different approaches, including traditional job portals, university homepages, web scraping, NLP-based methods, search engines and APIs.

We not only identify the solution that most completely fulfills the requirements set, but also discuss the reasons for its effectiveness. In this way, we aim to provide a conclusive answer based on a thorough analysis, which ultimately provides a reliable guide for the development of a job search platform tailored to the specific needs of PhDs. A brief comparison of how each method performs with regard to the defined requirements is shown in the evaluation matrix in Table 2.9.

Req. / Approach	Job Portals	University Homepages	NLP-Based	Web Scraping	Search Engine	APIs
R1	★★★★	★☆☆☆	★★☆☆	★★☆☆	★★★☆	★★☆☆
R2	★★★★	★★☆☆	★★☆☆	★☆☆☆	★★★☆	★☆☆☆
R3	N/A*	★★★★	N/A*	★★★★	★★☆☆	★★★★
R4	★★★★	★★★★	N/A*	★★★★	★★★☆	★★★★
R5	★★★★	★★★☆	★★★★	★★★☆	★★★★	★★★☆
R6	★★☆☆	★★★☆	★★☆☆	★★☆☆	★★☆☆	★★★☆
∑	**18**	**17**	**10**	**16**	**17**	**17**

Table 2.9: Evaluation matrix for the State-of-the-Art

Job portals have been consistently ranked as top performers with an overall score of 18, primarily due to their Low-Effort (R1), Ease-of-Use (R2), and Data Refresh and User Preference Retention (R5) capabilities. Their design places a strong emphasis on usability and simplicity. However, they can lack depth, particularly in terms of data completeness (R4) and PhD-Centric Solutions (R6), due to the reason that these platforms are rather centered around the general job search process.

University Homepages, being the primary sources for academic positions, naturally excel in Data Completeness (R4). Nonetheless, these sites frequently lack user-oriented features and receive low grades in areas such as Ease-of-Use (R2) and Data Refresh and User Preference Retention (R5), which results in a good score of 17.

NLP-based approaches possess plenty of potential but suffer from incongruities, which results in the lower score of 10. Their technical abilities face obstacles in Data Completeness (R4). Their efficiency typically depends on the quality of the source data.

Web scraping, although versatile which reflects in the score of 16, is constrained by the intricate nature of the process. It excels at Versatile Data Integration (R3) and data completeness (R4) but falters in delivering a personalized user experience (R2).

Search Engines and **APIs** have their own strengths and weaknesses. While search engines excel in data refresh and user preference retention (R5), they may not always offer the most specialized or comprehensive data for academic job searches. On the other hand, APIs provide high flexibility and complete data but may lack user-friendliness (R2).

In summary, although each technique has its own advantages, an all-encompassing answer is difficult to find. As we move towards developing a comprehensive job search platform specifically for PhDs, we may need to combine aspects of different strategies to find the right balance between depth, user-centricity, and scope.

2.5 Summary

In this review of job search methodologies, we have evaluated various approaches and their suitability for PhD job seekers. Our analysis indicates that while several platforms are suitable for general job searches, they lack the necessary nuance to cater to the academic job market. This disparity is even more evident for PhDs searching for appropriate professorships.

Our analysis emphasizes the need for a comprehensive platform that caters to the unique needs of PhDs. The various state-of-the-art methods display diverse performances, suggesting a potential solution of combining different approaches to create a tailored job search platform for PhDs.

In the next section of this thesis, our focus is on formulating a solution that incorporates the identified strengths from our state-of-the-art analysis while also addressing the identified gaps and creating a more streamlined job search experience for PhDs.

3 Concept

This chapter presents the concept for *ProfJobs*, a practical solution designed to help postdocs in their search for a professorship. Building on the previously discussed job search techniques, we propose a plan that synthesizes the best practices and addresses the shortcomings of current approaches. At the center of our solution is web scraping, a technique for collecting diverse data from web sources that provides a holistic view of the academic job market. We will look at the various aspects of this concept, from the technological standpoint to the user interface, and finally create a concrete, plan for an effective job search.

3.1 Overview

Our concept, illustrated in Figure 3.1, combines web scraping, data science, and Natural Language Processing (NLP) to automate the sourcing, preprocessing, and organization of academic job postings, specifically for professorial positions.

The following sections offer an examination of Phases 1 through 7 (see Figure 3.1), highlighting the role of each phase in transforming raw data into an organized and insightful compilation. This part of our concept will show how these phases function synergistically as the core of the proposed solution, laying the groundwork for an advanced platform tailored to the academic job market. Phases 1 and 7 involve manual processes but do not include implementing a solution. Instead, these phases concentrate on data provision and evaluation.

Evaluation, or Phase 7, is a critical element of this thesis and will be discussed in Chapter 5. This phase is dedicated to evaluating the effectiveness and impact of the proposed approach. The assessment will determine if the created solution meets the expectations and needs of its intended audience, and its usefulness in the academic field.

3.2 Data Source Identification

The identification of a data source is a crucial initial step in the research process, serving as the foundation for the entire solution. Our approach begins by identifying a multitude

of publicly available data sources within the D-A-CH region. A diverse array of data sources from different domains is selected, as it has the potential to enhance the rigor and resilience of the data.

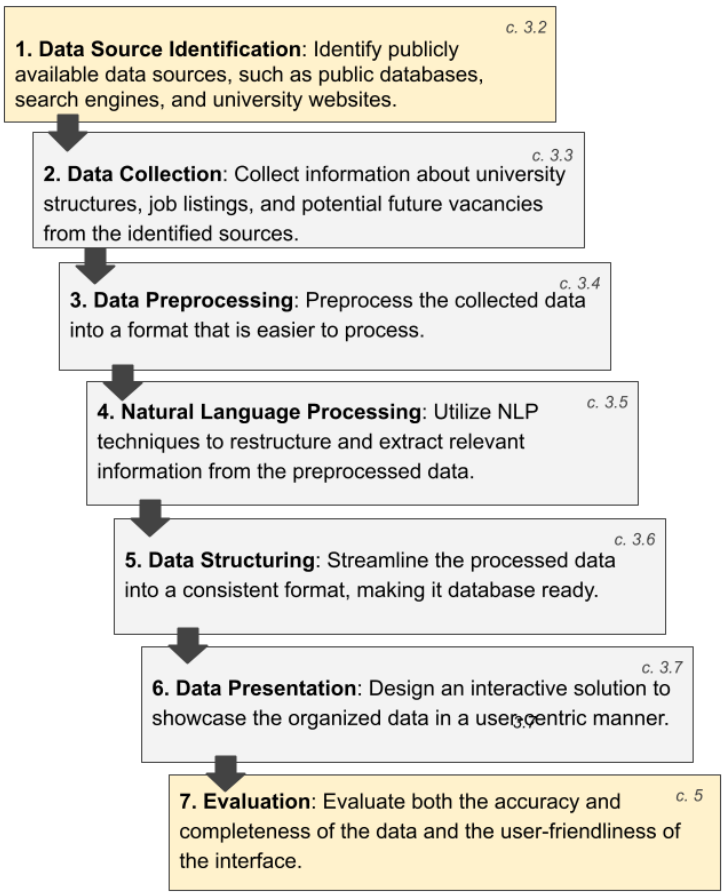

Figure 3.1: Concept: Overview of the involved steps

However, this broad range also introduces challenges in information retrieval due to the unique characteristics of each source. Furthermore, the integration of a data source that effectively maps the distinctive institutional structures of universities is crucial. This entails exploring various sources such as public databases, search engines, individual university websites, among others. These sources offer a spectrum of information that can be easily extracted through advanced web scraping techniques.

For the selection of our data sources, we have followed a set of specific criteria to guarantee the accuracy, relevance, and comprehensiveness of our solution:

- **Reproducibility:** Emphasizing reproducibility is at the forefront of our criteria. We prioritize publicly available sources to enable the replication of our research by other scholars and enthusiasts in the field.
- **Exclusion of Restricted Content**: We consciously avoid sources that have restricted access due to privacy constraints or pay-walls. Our solution is based on transparency and accessibility, ensuring that our findings are not reliant on information that is not readily available to the public.
- **Web Resource Orientation**: Our approach, *ProfJobs*, is focused on web scraping, which directs us towards web-based resources. These resources provide a vast amount of information and allow for automated data extraction and processing.
- **Recency and Update Frequency**: Public web resources have an advantage in frequent updates. The dynamic nature of the web ensures that the extracted information is current. This provides a timely and relevant snapshot of the academic landscape including professorial job offerings.

Given the outlined criteria, we have segmented our data sources into distinct categories, each providing essential information for our research objectives:

- **Public Databases:** For example, the DFG GERiT database[11] is a robust public resource that provides a comprehensive directory of research institutions throughout Germany.
- **Semantic Web Databases:** An example of this category is DBpedia[12], a resource that obtains structured content by extracting information from Wikipedia to provide enriched, contextual data.
- **Search Engines:** Search engines like Google and Bing serve as broad-based data collectors, scraping the internet and compiling a wealth of information, making them an important source for our data collection.
- **University Homepages:** Specific university websites, such as the TU Chemnitz homepage[13], are rich resources that provide detailed insights into academic job listings, faculty composition, current professorships, and ongoing research.
- **Job Portals:** As explained in the previous chapter job portals serve as primary resource for job listings. In addition to university-specific platforms, these general websites offer a wide range of job listings, including in the academic field.

[11] https://www.gerit.org/de/
[12] https://www.dbpedia.org/
[13] https://www.tu-chemnitz.de/

During our investigation into potential data sources for *ProfJobs*, we intentionally opted against certain approach groups. Specifically, we chose to abstain from utilizing job search application programming interfaces (APIs) due to the inherent cost and lack of public availability. These constraints could present challenges for other scholars wishing to reproduce our research. Similarly, existing NLP-based solutions were not suitable for our needs as we found that many were designed specifically for a recruiter's perspective or for general job searches, and a significant number of these strategies were not publicly available. Moreover, some required additional usage fees and, crucially, lacked the ability to export data.

Category	**Example**	**Primary Purpose**
Public Databases	DFG GERiT	Mapping out the structure of academic institutions
Semantic Web Databases	DBPedia	Providing structured content with enriched, contextual data
Search Engines	Google, Bing	Broad overview of job postings and general data collection
Job Portals	Academics.com	Primary source of academic job listings
University Homepages	TU Chemnitz	Detailed insights: academic offerings, faculty specifics, ongoing research

Table 3.1: Identified Data Sources for Academic Job Landscape

In summary, the identified data sources, as shown in Table 3.1, are crucial for the success of our solution. The DFG's GERiT database plays an integral role as it provides a structured overview of the German research environment, making the mapping of complex and diverse academic structures more manageable. Search engines, along with accessible websites such as job portals, are essential for a broad range of academic job offers in the D-A-CH region. For the practical demonstration of our approach, we will primarily use the academics.com job portal. Our manual investigation showed that, at the time of writing, it has the most comprehensive listings for professorships among the portals we examined (see 2.2.1).

University websites are also crucial, as they provide detailed information about job offerings, faculty composition, and ongoing research activities. This information is crucial for our solution because it contributes to a more nuanced comprehension of academic

job listings. Together, these data sources establish a strong and diverse basis for *ProfJobs*, allowing us to automate the collection and arrangement of job postings, with a focus on professorial positions.

3.3 Data Collection

The data collection phase of our solution involves a methodical information extraction process from the previously identified data sources. For example, a specific task may require retrieving HTML content from a university's official site. It is crucial to acknowledge that present-day websites are highly dependent on technologies such as JavaScript for displaying dynamic content, necessitating the use of techniques to render the JavaScript in order to guarantee the full data access. [32]

Scraping information from the web presents not only technical challenges, but also ethical challenges, particularly when it affects individuals' privacy rights. Committed to ethical principles, our solution adheres to the standards established by the General Data Protection Regulation (GDPR) in the European Union. This approach reinforces our dedication to preserving individual privacy rights [33]. Additionally, following the web scraping guidelines presented by Mazilu et al. [34], our solution prioritizes ethical web scraping practices. This includes adhering to website-specific directives, such as the robots.txt file, and to prevent our data retrieval mechanisms from overwhelming servers with excessive requests. Additionally, our web scraping tools will incorporate metadata, when possible, to clarify their purpose and provide contact information via the scraper's useragent string, in the interest of transparency and collaboration.

In the following sections we introduce the specifics of our selected data sources and the reasoning behind each choice.

DFG GERiT. Mapping the academic terrain of the D-A-CH region is the first step in our data collection, and this is where DFG GERiT proves to be superior. It is a comprehensive repository rich in insights about the structural information of over 29.000 universities and research institutes in Germany. It contains point to specific web pages of professorships. The database of DFG GERiT is available in Excel format, which is updated monthly, allowing for easy extraction and analysis [35]. The Excel version, while user-friendly, is scarce in terms of data points. It does not comprehensively reflect the web

catalog. Given this limitation, our strategy is to scrape the GERiT's web catalog directly instead.

We utilize web scraping techniques to automatically navigate through the hierarchical structure of the web catalog, to systematically fetch the HTML of the individual pages. In this way, comprehensive information can be extracted about the universities, the institutes that belong to them, and the numerous professorships they offer. At this stage, the URLs associated with each of these institutions are also collected. To further enrich this dataset, the web catalog provides descriptions about the universities, complemented with a visual representation of the university. With respect to the data presentation phase of our project, such content and images are invaluable and will also be collected.

It is crucial to acknowledge another limitation of DFG GERiT: its primary focus on the German academic field. Our solution aims to encompass not only Germany but the entire D-A-CH region. This brings us to our next data source to fulfill our regional focus and also map the academic landscape of Austria and Switzerland.

University Homepages. University homepages serve as important gateways to the academic world. They not only provide access to the individual homepage of professors', but also critical information about universities and research institutes, including up-to-date and precise details. These homepages are more than just repositories of information; rather, they serve as primary displays of an institution's academic and research landscape.

Drawing a parallel with job portals reveals that data extraction from university websites often requires the use of web scraping techniques. Given the prominence of JavaScript in contemporary web design, it is imperative that any scraping tool has the ability to render this scripting language. This enables a comprehensive data extraction. Nevertheless, the obstacles that university homepages present are multifarious. There is considerable diversity in the content structure not only among different institutions but also within departments or faculties of the same university. The presentation of job positions is a clear example of such heterogeneity, as it can vary widely in terms of display and content depth.

As a solution to these complexities, a robust and flexible web scraper is required. A proficient tool for effective content management must also excel in the execution and rendering of JavaScript.

Job Portals. Job portals are essential to our data collection strategy as they serve as rich repositories of current job openings at universities. The vast information offered by this data source makes it invaluable. Nonetheless, it is important to note that it is often not available in structured formats like public databases.

Some job portals provide structured access to their data through Application Programming Interfaces (APIs). This enables precise data retrieval in a structured format, suitable for direct integration into our database. Nevertheless, APIs have their limitations, such as being frequently non-public and reserved for HR departments or partner firms and incurring additional usage-based costs. Due to these limitations and our commitment to accessibility, we have not used this method as our primary data collection approach.

Instead, given the challenges of direct API access, we decided to use web scraping to extract job postings. This technique enables us to access a wide range of job opportunities listed on the platforms, despite the absence of structured data. The challenges posed by university homepages are similar to those encountered with job portals. Most job portals heavily depend on JavaScript to dynamically populate their pages with search results, which requires a scraper capable of evaluating and rendering JavaScript. When we take a look at scraping large volumes of data, it is crucial that the scraper is resilient. Given that job portals aim to uphold site integrity and prevent misuse, they frequently employ advanced bot detection mechanisms to prevent the scraping.

Given these challenges at hand, it is recommended to utilize browser-based tools like Puppeteer or Selenium. These tools, when operating in *'headed'* mode, where the active browser interface is visible, can greatly decrease the chance of bot detection triggers. For Puppeteer, a plugin system is available that includes a stealth plugin. This plugin reduces the likelihood of the browser being identified as headless and therefore a bot by emulating various APIs, such as spoofing a GPU.

Moreover, to ensure the continuity of data collection and conceal our scraping activities, implementing a few additional measures is imperative.

- **Rotating IP addresses:** We implement IP rotation across various subnets and distinct geographical locations to make the scraper's actions seem more like natural web traffic.
- **Rate Limiting & Sleep Intervals:** The timing of data collection process is also critical. Introducing sleep intervals during page crawls and establishing rate limits not only prevents hitting server-imposed restrictions but also avoids overwhelming the server with requests, potentially resulting in IP bans or temporary server downtimes.

Together, these strategies ensure that our solution can scrape data from job portals without being limited or blocked.

Summary. Summarizing the data collection process reveals that every data source presents distinct challenges, demanding customized strategies for optimal information retrieval. Web scraping is beneficial for DFG GERiT, job portals, and university homepages. This diversity highlights the necessity for a multifaceted approach to secure thorough and precise data retrieval of our solution, while also keeping the ability to automate the data collection in mind.

3.4 Data Preprocessing

After collecting data, it is essential to convert the raw information into a more analyzable format. Raw HTML-based datasets often contain excessive details that have no relevance. Hence, data preprocessing is necessary to eliminate such irrelevant information and derive more purposeful insights. This phase is crucial when preparing data for NLP applications. An optimal preprocessing routine guarantees that our NLP models receive high-quality data, which is necessary for the success of any language processing task.

There are various techniques with distinct benefits and drawbacks for the preprocessing:
Manual Inspection and Processing: Manual inspection involves a detailed analysis of the data line by line. This process demands a keen eye for detail and a deep understanding of the context surrounding the data. The result is an accurate and cleaned dataset that can greatly benefit complex NLP tasks where nuances are significant. Despite its precision, this approach is rarely used exclusively due to its time-consuming nature and the high labor demands it places on humans.

Regular Expressions. Regular expressions or *regex* are a valuable tool for identifying and extracting patterns in text. They work well with both structured and semi-structured data but require a significant initial investment to develop precise patterns to avoid false positives or false negatives. The inherent fragility of *regex* means that changes to the source structure can make previous patterns obsolete, necessitating continuous maintenance.

Metadata Usage. Metadata, specifically employing schema.org[14] annotations, allows HTML content to be semantically tagged for easy machine reading. For job portals, metadata serves as a useful and readily accessible source of information, because most of them are designed to optimize their listings for search engines. The consistent format streamlines the process of extracting information, enabling retrieval of well-defined data blocks. However, relying on metadata assumes that the source maintains precise and comprehensive labeling, a condition that is not always fulfilled.

Software Libraries. Software libraries provide an intermediary solution between manual inspection and full automation by supplying functions that target HTML elements, attributes, and text in the extraction process. For instance, BeautifulSoup [36], a popular Python library, excels in navigating and searching the parse tree of HTML. However, effective usage of them demands programming expertise and acquaintance with HTML architecture. SmartReader [37] which is based on the Mozilla Readability library[15] concentrates on excluding nonessential components, such as navigation and ads, to extract the primary content from web pages. Likewise, Textify [38] is a package designed to convert HTML content into a text-based format while efficiently disregarding irrelevant or minimally informative elements.

In practice, our approach to preprocessing data is not dependent upon a singular method. Instead, we opt for a hybrid approach that is tailored to the specific nature of the data source being analyzed. For complex or unstructured datasets where accuracy is critical, manual inspection may be indispensable. Conversely, for more structured and standardized data, we can take advantage of the efficiency provided by regular expressions and metadata parsing. While regular expressions and software libraries provide scalability, they occasionally need fine-tuning to accommodate structural changes in data sources.

3.5 Natural Language Processing

In the NLP phase of our concept, the structured data undergoes thorough review and refinement to highlight relevant information. This phase uses advanced NLP technologies, especially Large Language Models (LLMs), to distill and interpret complex details.

[14] https://schema.org/docs/documents.html
[15] https://github.com/mozilla/readability

Due to the limited input capabilities of modern LLMs, it is necessary to thoroughly clean the data to optimize their performance, which was done in the data preparation phase. When it comes to accessing and using LLMs, there are several options to take, each with its own set of benefits and challenges:

API-based: APIs for LLMs, such as those from OpenAI or Claude, provide an accessible entry point for users. They have the advantage of being easy to use and require minimal programming skills due to the comprehensive libraries available for all major programming languages. The trade-off for this ease of use is the cost of processing, which is proportional to the amount of data being input and output.

A special type of these is OpenRouter (OR). It provides a common interface for LLMs that supports a variety of models from leading organizations such as OpenAI (ChatGPT), Meta (Llama), and Claude. At present, it supports 43 models, with select beta models accessible for free for a limited duration. Despite its versatility, it has a shared drawback with other APIs: the processing costs of most models, specifically non-beta ones. [39]

Self-Hosted: The alternative to using APIs is to use an open LLM like LLaMA [40], LLaMA 2 [41] or Alpaca [42], which are available for free and offers a lot of flexibility due to the existence of fine-tuned models for different purposes. The gap between these models and proprietary counterparts such as OpenAI's GPT-3.5 has been narrowing, with open-source options showing significant progress in the recent months [43]. The main drawback is resource intensity: running such models requires significant GPU memory and processing power, and open-source models still tend to be outperformed by closed source models such as GPT-4.

A clear advantage of self-hosting models is their ability to be adaptable to specific objectives and data sets. Unlike proprietary models, these models can be customized, trained from ground, or fine-tuned to meet unique requirements. [44] To provide context about the costs, each training run of OpenAI's GPT-3 for example is estimated to have cost over $4.6 million, suggesting the process involved multiple iterations. During an event at the Massachusetts Institute of Technology (MIT), a co-founder of OpenAI revealed that the cost of training foundational models like GPT-3 amounts to more than $100 million in total. This highlights the significant investment required for developing state-of-the-art language models from the ground up. [45]

Nevertheless, as of this writing, fine-tuning such models, which does not involve constructing the model from scratch, remains an impractical task for most users. The main obstacle is the requirement for up-to-date, high-performance GPUs, ideally those utilized in datacenters, to support the demanding processing power and memory needs of these models. This prerequisite presents a substantial limitation to the practicability and attainability of fine-tuning these cutting-edge language models for individual users or small-scale firms. However, at the time of writing, there have been numerous contributions from the FOSS community, particularly on GitHub, aimed at enabling the optimization of these models on high-end consumer GPUs using lower precision float calculations, thereby reducing the resource demands [46]. Therefore, while self-hosting provides certain benefits in terms of customization and flexibility, it also presents significant challenges, especially when it comes to the necessary computational resources.

Cloud GPU Platforms: Platforms such as Google CoLab[16] offer the ability to host smaller LLMs on cloud-based GPU services for free. However, server resource limitations can throttle performance or even shut the free service down when demand is high, making such solutions ineffective for processing data at the scale we need.

Free Web-based Versions: There are free editions of prominent LLMs, such as GPT-3.5 or Claude, that offer limited request quotas. These quotas are dynamically adjusted according to the current availability of computing resources. The downside of these free versions is their limited accessibility, which typically requires manual interaction which hinders automated, large-scale data processing. Another drawback is that Claude is only available in the U.S. at the time of writing.

Subscription-Based: Subscription-based models for accessing LLMs, such as OpenAI's GPT-4, provide a powerful solution for organizations that need reliable and scalable NLP capabilities. These models are distinguished by their ability to process larger volumes of data than their free counterparts. Subscribers benefit from predictable pricing structures that make it easier to budget and plan for regular usage. However, these benefits come with trade-offs. The recurring cost of subscription services can be a significant factor for smaller organizations or projects. While subscription models offer higher resource limits, they can still limit larger or automated tasks because they can only be

[16] https://colab.research.google.com/

accessed through the web interface. There are also privacy and security concerns, as sensitive data is often processed and stored on third-party servers.

Based on an objective assessment of the available access methods and their respective advantages and disadvantages, we determined that OpenRouter[17] with its free beta models and extended library of more advanced models is the most practical and cost-effective solution for *ProfJobs*. Our decision was based on the limitations of free versions, such as ChatGPT, and the impracticality of self-hosting, which requires significant local hardware resources. OpenRouter offer us strong processing abilities and the flexibility of cloud-based resources, eliminating the need for extensive local infrastructure. This strategic decision aligned to our solution's demands for efficient and scalable NLP processing, while also considering budgetary constraints.

3.5.1 Vacancy Estimation

The process of forecasting academic staffing needs for PhDs relies heavily on accurately determining the vacancy date of a professorship. To achieve this, it is essential to determine the age of the professor in question. Through our investigation of the faculty of computer science at TU Chemnitz, we discovered that professors' birthdates are rarely explicitly stated on their web pages. However, these pages frequently include additional information, such as CV details or lists of publications, that could be utilized to deduce their age. We suggest implementing NLP techniques to extract this data.

For accurately estimating the age of a professor, based on available CVs or publication lists, a carefully crafted NLP prompt is necessary. This prompt must provide both contextual information and specific instructions to guide the model towards generating the desired output. Furthermore, considering the capability of using system-level prompts with OpenAI's API, we propose employing such prompts at the system level. This approach assigns a higher weight to these prompts compared to user-level prompts, thereby more effectively directing the model to adhere to a specified output format.

Estimating the potential vacancy or retirement of a professor requires evaluation of their age and the typical retirement age. In Germany, professors commonly retire between the ages of 63 and 65. Notably, the average age of professors in Germany is on the rise, emphasizing the relevance of predicting future vacancies. In 2020, Germany had 8,010

[17] https://openrouter.ai/

professors aged between 60 and 65, and 2,165 professors over 65 years old. This demographic trend emphasizes the importance of creating a method for estimating vacancies. [47], [48], [49]

The upcoming number of retirements additionally emphasizes this need. Projections predict that approximately 14,992 professors in Germany are expected to retire from 2024 to 2030. This presents significant opportunities for aspiring PhDs. [49] [50]

To calculate the retirement age, we have set two boundaries: a lower limit of 63 years and an upper limit of 65 years. Applying these boundaries, we have developed two equations to predict the lower (3.2) and upper (3.3) limits, based on the common equation (3.1), of when a professorship is expected to become vacant. The equations are as follows, where CY denotes the current year, A is the estimated age of the professor, and R is the retirement age:

$$\mathrm{V} = \mathrm{CY} + (\mathrm{R} - \mathrm{A}) \tag{3.1}$$

$$V_{\mathrm{from}} = \mathrm{CY} + (63 - A) \tag{3.2}$$

$$V_{\mathrm{to}} = \mathrm{CY} + (65 - A) \tag{3.3}$$

The formulas presented here offer an estimate for when a professorship position may become vacant, based on the age of the current professor and standard retirement ages. Though the methodology relies on projections, it provides a systematic approach for estimating potential future vacancies on the professor's age.

3.5.2 Parsing Job Listings

An essential part of our NLP solution involves gathering and converting job listings into a structured format. This structured data is then used in our solution to present job opportunities in a more accessible and user-friendly manner. The use of LLMs for this task represents a substantial improvement over conventional means such as regular expressions.

Traditional methods, such as using regular expressions, necessitate developing customized patterns for every website or platform's individual arrangement and structure. This technique is a demanding task that lacks versatility. Even minor changes to the website's structure need corresponding modifications to its unique code expression. The process is prone to errors and takes a considerable amount of time, particularly when handling a diverse range of job listing sources.

The utilization of LLMs offers a dynamic and flexible methodology. Custom prompts guide the model in comprehending the environment, goal, and required output format. These prompts instruct the model on extracting and structuring details from numerous job listings. A system-level prompt proves advantageous in this scenario, as it gives the prompt more weight than the actual input data.

Using LLMs to parse job listings has several advantages:

- **Adaptability:** LLMs can adjust to diverse formats and structures of job listings on various websites and platforms without requiring personalized pattern writing, as required with regular expressions. [51]
- **Efficiency:** Updating and maintaining parsing capabilities is streamlined through adjusting prompts instead of rewriting complex regular expressions for each website change.
- **Accuracy**: LLMs can accurately interpret and categorize information in job listings, even if they are unclear or contain complex language, because of their advanced understanding of language and context.
- **Scalability:** This approach is inherently scalable because the same model can be applied to many sources with minimal customization, making it ideal for an application that aggregates large volumes of job postings.

In summary, implementing LLMs for parsing job listings in our web application offers a more efficient, accurate, and scalable solution compared to traditional methods. This approach not only improves the user experience by providing more reliable and well-organized job information but also optimizes the backend processes, resulting in a more robust and easily maintainable system.

3.6 Data Structuring

After concluding the collection and preprocessing phases, a crucial next step is to organize and incorporate this data into a layout that is easily understood and readily available. Not only does this incorporation ensure seamless storage and handling, but it also streamlines the process of data retrieval.

Databases are the primary method for this objective. Relational databases are highly effective for datasets with clear relationships, due to their structured schemas and powerful querying capabilities. Scalability may pose occasional challenges for extremely large datasets. Non-relational databases, on the other hand, excel in flexibility, scalability, and

handling unstructured or semi-structured data. However, they may not provide the same level of in-depth querying capabilities as their relational counterparts. [52]

Additionally, various formats and techniques can fulfill this purpose beyond traditional databases. File-based storage provides a simple and transferable solution, particularly when working with immediate data retrieval that does not require a database management system. Excel spreadsheets are a user-friendly option for manual data entry and visualization. However, they may not be suitable for very extensive datasets.

JSON and XML have gained popularity due to their flexibility and wide acceptance in today's digital age. JSON is known for its lightweight nature and easy readability and writing. On the other hand, XML offers a structured data representation capable of managing complex structures and relationships. Both formats cater to multiple programming languages and facilitate seamless data parsing. [53]

Our solution employs non-relational databases due to several advantages they provide. First, their inherent flexibility enables seamless handling of diverse data formats collected from various sources, which is crucial since our data sources are heterogeneous. Second, their scalability guarantees the robustness and efficiency of our system in the face of potential dataset growth and expansion. Lastly, their proficiency in managing large amounts of unstructured data aligns with our aim to provide a comprehensive academic overview without rigid frameworks. [52]

Choosing the appropriate storage method is crucial for data structuring, considering the data's characteristics and meeting the end-user's needs. An efficiently structured dataset is the foundation of a user-friendly interface, enabling comprehensive data view and functionalities such as searching, sorting, and filtering.

3.7 Data Presentation

To address the previously identified issues, it is paramount to have a user-friendly presentation of the data. No matter how extensive or comprehensive the data is, it can only serve its purpose if it is presented in a clear and actionable way. Therefore, our primary objective during this phase is to prioritize user accessibility and interaction.

Choice of Platform. We deliberately favor a web-based solution because standalone applications can require local updates and have been limited to specific operating systems,

which prevents easy and widespread usage. Conversely, web-based platforms offer universal access, allowing users to access our services via a variety of devices such as desktops, tablets, and smartphones, providing the necessary flexibility. Nevertheless, web-based platforms have their challenges too. Poor network connectivity in some regions limits access, and browser disparities may lead to inconsistent user experiences.

Embedded Functionalities. Including features like favoriting, filtering, and exporting data is vital for modern web interfaces. A static page with plain listings would be overwhelming and off-putting when compared to the necessary user engagement and interactivity which allow for a personalized experience. By providing a tailored environment, users can identify job opportunities that align with their aspirations without sifting through irrelevant listings. Still, there are costs associated with these functionalities. The system's backend infrastructure must be sturdy, and user data security is always a worry, particularly when user data is saved.

Design and Accessibility. In a world that emphasizes inclusivity, platforms that do not adhere to accessibility guidelines run the risk of alienating a significant portion of their audience. While some platforms offer accessibility as an add-on or an afterthought, our approach ensures a consistently accessible experience. While flashy visuals may be given priority in modern designs, we recognize the significance of a balanced design. By prioritizing elements such as contrast and avoiding potential accessibility barriers, we not only cater to those with disabilities but also ensure an enjoyable user experience for everyone. [13]

Colors and Aesthetics. The selected color scheme, combining modern and professional elements, was not only an aesthetic decision. The color selections were influenced by the research of Hallock et al. [54] and Kondratova et al [55]. Their findings provide insightful information regarding the psychological and cultural effects of colors. Hallock et al.'s research identified distinct associations connected to different colors: blue connotes trust, security, and dependability; yellow and orange frequently convey affordability; and red and black provoke feelings of fear while red also implies speed. Black is often associated with high technology and quality. These associations can significantly impact user perceptions and emotional responses towards our platform.

Kondratova et al. [55] adopted a distinct strategy by creating an international color scheme that mainly features white, gray, black, blue, and yellow. This color palette proves to be highly effective in catering to a global audience as it eliminates cultural barriers and is generally well-received in various regions.

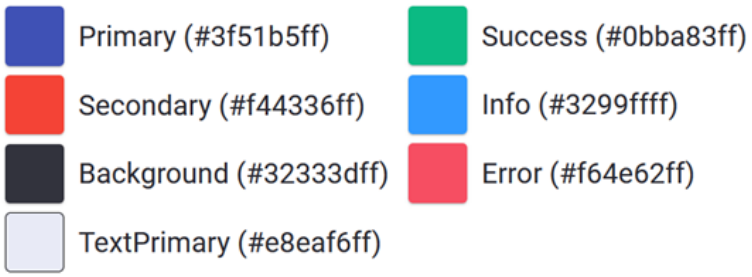

Figure 3.2: Color Palette for ProfJobs

When creating our color scheme, we drew inspiration from these studies to guarantee that our website appeals to a globally diverse audience while embodying attributes such as trust, security, reliability, as well as an association with advanced technology and quality. These characteristics are imperative for a platform aiming to establish academic credibility and technological advancement.

The results of our research have been reflected in the color palette presented in Figure 3.2. This palette constitutes a thoughtfully selected combination of colors that aligns with our objective of developing a web interface that is both visually appealing and psychologically and culturally inclusive. Through this thoughtful selection, we aim to foster an environment that is welcoming, trustworthy, and indicative of the high-quality content and services our platform provides.

Wireframe Design. The wireframe design of our platform, shown in Figure 3.3, serves as a blueprint, providing a visual guide to the intended user interface of our job search solution. The design was created with a clear, user-friendly layout in mind, with the goal of ensuring an intuitive navigation experience for our academic audience.

At the top left of the interface is the placeholder for our logo, symbolizing the brand identity and mission of our platform. In the center of the page is a menu bar that provides navigation with links to essential pages such as Home, Favorites, Institutions, and a novel feature that estimates potential future job opportunities.

Along with the search bar on the left side of the page, we present the primary filtering capabilities. Here, users can finetune their search results based on a variety of criteria, including country, posting date, university, department, and salary level. This section is designed to help users customize their job search to meet their specific academic career needs and preferences. The central panel of the interface displays job listings dynamically. The listings are briefly summarized, presenting crucial information including job title, university name, country, city, and an institution image (where applicable).

Additionally, every listing includes a short description providing a brief overview of the position.

Figure 3.3: User Interface Wireframe Design

Inspired by the established job search portal StepStone[18], our wireframe is designed to embody a proven user-friendly and efficient layout. By emulating the best aspects of these industry leader, we strive to create a familiar yet specialized environment that meets our users' expectations and enhances their academic job search experience.

3.8 Summary

In previous chapters, we introduced our approach to simplify the academic job search, prioritizing the needs of professors, postdocs, and senior PhD students. It encompasses multiple phases fundamental to the solution, commencing with the identification of potential data sources in Chapter 3.2, followed by the collection of that data in Chapter 3.3. Recognizing the importance of data size, we delved into the process of data

[18] https://www.stepstone.de/

preprocessing to transform raw data into a refined format more suitable for further analysis in Chapter 3.4.

Subsequently, we explored the advanced domain of NLP. Leveraging state-of-the-art LLMs, for extracting information from the pre-processed data to get insights that might otherwise stay concealed in unstructured text in Chapter 3.5. The ability to predict the retirement age of professors through CV or profile page analysis and the parsing of job listing with NLP proved to be a notable feature of our solution. In Chapter 3.6 we introduced to the structuring of the data and the reasoning behind it. The last Chapter 3.7 introduced to the data presentation and how that structured data can be used to display it in a user-friendly format.

This concept lays the groundwork for the upcoming implementation of our solution, where we explain how all parts combine to produce an effortless and user-friendly system. Our objective is to equip job seekers with the essential tools needed to locate appropriate positions and offer them a preview of foreseeable employment opportunities.

4 Implementation

This chapter focuses on implementing a solution designed to enhance the academic job search for postdocs and represents the transition from the conceptual design to practical implementation. The processes and technological choices that support them are described in detail, and the technical implementation of our proposed system is presented.

This chapter methodically breaks down each step of our approach, from data collection to the final presentation within our functional prototype. This prototype is a concrete manifestation of our research efforts, designed to meet the unique needs of professors, postdocs and senior PhD students searching for professorial positions. By the end of this chapter, readers will have a complete understanding of the prototype's development and its usefulness as a specialized platform for academic job searches.

4.1 Overview

In this chapter, we offer an overview of the key phases involved in implementing our proposed solution for enhancing the academic job search process for PhDs. Each phase plays a critical role in converting the raw data into a cohesive, user-friendly platform that streamlines the process of finding vacant professorial positions. The process of transforming raw data is illustrated in a BPMN diagram in Figure A.1.

Initially, we will examine the design and function of the web scrapers developed to collect data from specified sources, as detailed in our proposal. Followed by the phase of data preprocessing, where we will refine the collected HTML data to eliminate irrelevant information and retain only that which is essential for our objective. After, we will proceed with the Natural Language Processing (NLP) phase, where the goal is to extract structured, meaningful content from preprocessed, unstructured data. The subsequent structuring phase will organize this content into a standardized format, ensuring uniformity.

The last phase focuses on how we present the structured data through an interactive web interface that specifically addresses the needs of PhDs in their search for professorial positions. We will describe how this platform not only presents the collected data and job listings but also offers a personalized journey that simplifies the job search process. Throughout the implementation, Git was utilized for version control to ensure

systematic tracking. To guarantee that our solution meets the needs and expectations of its intended users, Chapter 5 evaluates the effectiveness and impact of the implementation.

4.2 Data Collection

The data collection chapter of this thesis provides a fundamental framework for harvesting information across diverse web-based platforms in a systematic manner. It is important to acknowledge the multifaceted nature of our approach, which includes a range of sources such as academic databases, search engines, job portals, and university homepages. Each source requires a customized approach for effective navigation of its distinctive terrain, guaranteeing the precise and ethical extraction of data. This chapter explains the methods and technologies used to collect a dataset that will serve as the cornerstone for subsequent analysis and presentation.

DFG GERiT. For scraping the DFG GERiT database, we created a web scraper utilizing C# programming language, in conjunction with the .NET 8 framework. Our selection was based on our conceptual phase discussions, during which we found that the Excel database provided by DFG GERiT contained only a fraction of the data available in the web catalog. C# was picked due to its ease of programming, extensive tooling, documentation, and the rich library of built-in functionality that .NET provides. Notably, the HTTP Request class in .NET helped to efficiently make asynchronous HTTP requests to the web server.

Our preference for .NET 8 over the traditional .NET Framework was based on two reasons. Firstly, it can operate on different operating systems, such as Windows, Linux, or macOS, which broadens its scope of use. Secondly, since the traditional .NET Framework is reaching its end of life, Microsoft recommends .NET 8 for new applications, ensuring that our tool remains up-to-date, future-proof and profiting from the latest improvements and performance optimizations. [56]

During development of the scraper, examination of the website's HTML determined that the page content was dynamically loaded via JavaScript from a JSON object. For efficient data capture, our scraper was designed to identify and locally store this JSON object for later processing. Subsequently, the scraper was used with all 1893 institutes listed at the time, enabling comprehensive information collection about each institute, including faculties and professorships. In total we collected the JSON object from 1873 out of the 1893

institutes. We were unable to scrape the data of 20 institutes because these were not found on the web catalog using the ID from the downloaded database. The entire implemented data collection process is illustrated in Figure 4.1.

Compliance with ethical web scraping practices was crucial throughout the process. Before scraping, we reviewed the robots.txt file of the DFG GERiT website to ensure our actions aligned with their restrictions, as described in our concept chapter. Additionally, to promote transparency and facilitate open communication channels, our scraper's requests contained our contact information alongside with the product name *ProfJobs*, in the HTTP useragent. This enables DFG GERiT to quickly identify the traffic source and can contact us if necessary. To avoid overloading the server, we added a one-second delay between requests. This minimized the risk of server overload, ensured server stability, and adhered to correct web scraping practices.

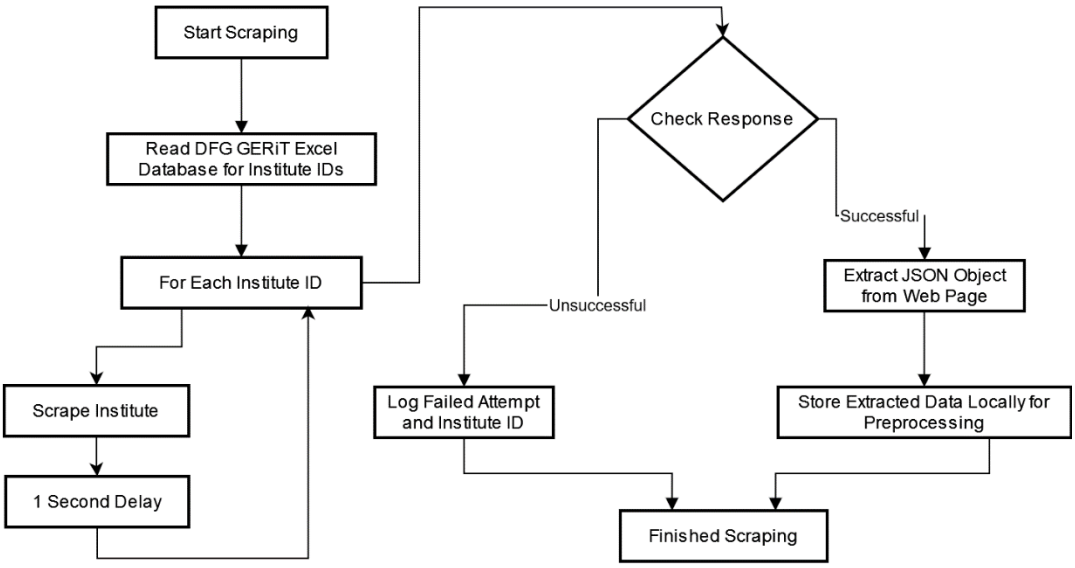

Figure 4.1: GERiT Scraper: Data Collection Flow

Website Scraping. To develop a reliable web scraping approach for university homepages, we acknowledged the dynamic nature of these web resources. JavaScript, which is fundamental to the interactive features of these sites, needed to be interpreted and rendered by our scraper. We focus on using browser-based scraping tools, such as Selenium and Puppeteer, to capture the websites content. These tools, initially designed for automating website and web application testing, also excel at web scraping by being able to replicate user interactions with web pages. To access JavaScript-generated content, their inherent capability to render web pages just like a standard browser is fundamental. [57]

To ensure consistency with the other parts of our data collection solution, we intended to integrate this scraping solution in C#. However, we reconsidered this approach after discovering the absence of stealth capabilities in both C# implementations of Selenium and Puppeteer. These stealth functions are crucial for circumventing detection measures, such as scraping and anti bot protection, as well as rate limits, that job portals usually use in order to prevent web scraping efforts.

As a result, we shifted to utilizing Node.JS with Puppeteer, which includes a community build plugin system called *puppeteer-extra*[19]. It also includes a stealth plugin, called *puppeteer-extra-plugin-stealth*[20]. This plugin facilitates our scraper's ability to emulate an authentic browser even though its running in headless mode. Additionally, we incorporated the *puppeteer-cluster* extension package to optimize our scraping process by utilizing parallel, isolated browser instances, thus accelerating the data collection through simultaneous acquisition from multiple web sites. To mitigate the risk of IP address blocking or rate limiting, we employed 100 proxy servers that were distributed worldwide. Each page will be scraped using the *puppeteer-page-proxy*[21] package, with the servers being dynamically assigned in a round-robin fashion.

To ensure accessibility within our C# environment, we encapsulated the web scraper with an express-based RESTful API. This allows us to initiate the scraping process through API calls from our C# codebase and consolidate our data collection efforts. Our final implementation of the web scraper and its components is illustrated in an components diagram in Figure 4.2. To improve manageability, we utilized a Docker image to containerize the web scraping component. The image includes the Node.JS-based scraper and its dependencies, streamlining the deployment process on any Docker-supported system without the need for manual dependency installations. Furthermore, this arrangement simplifies horizontal scaling by enabling deployment of multiple instances on separate virtual servers as needed.

[19] https://github.com/berstend/puppeteer-extra

[20] https://github.com/berstend/puppeteer-extra/tree/master/packages/puppeteer-extra-plugin-stealth

[21] https://github.com/Cuadrix/puppeteer-page-proxy

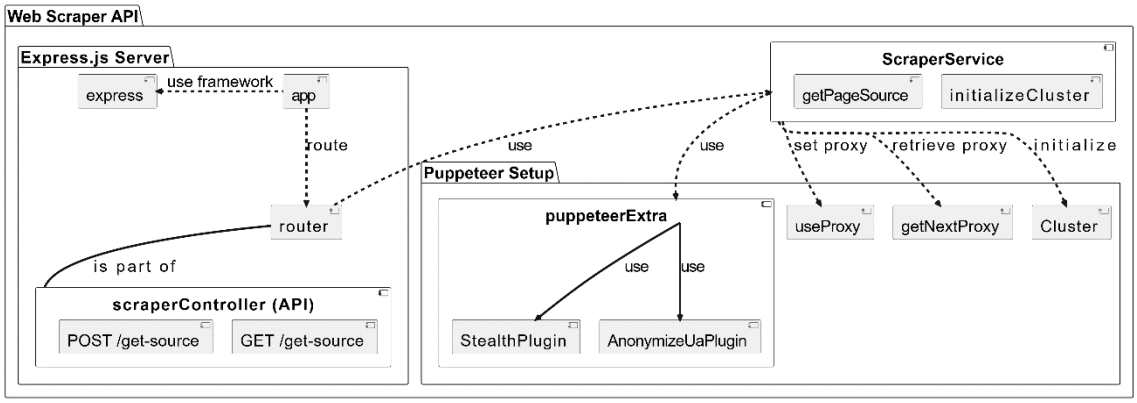

Figure 4.2: Visualization of the Web Scraper Components

Academics.com. For the extraction of data from academics.com, a dedicated web scraper was created to interact with two of the website's internal APIs. The process involved searching for new job postings related to professorships and using the keyword “professorship” as the primary filter. This method ensured the recovery of job opportunities relevant to the academic field, specifically catering to professorial positions.

After querying the initial data, the web scraper utilized the second API to gather detailed information about the listings. This was an important step for acquiring comprehensive data about each job posting, including specific details such as job descriptions, URL, application emails, locations, and application deadlines.

The scraper was implemented using C# with the .NET 8, chosen for its robust and efficient processing capabilities. The asynchronous scraping process, utilizing both Get and Post web requests, according to the APIs needs, enabled efficient crawling of the JSON formatted data. The final setup of the web scraper was installed on a Linux server, which was chosen for its reliability and low costs. To ensure consistent data collection, execution was automatically scheduled every 24 hours using the Cron-Scheduler, a time-based job scheduler in Unix-like operating systems. This automation ensured regular updates that improved the efficiency of the scraper and the relevance of the data extracted from academics.com.

In addition to the mentioned implementations for data collection, we have also created a concept and an implementation for scraping data from search engines, which can be found in Appendix A.5. Due to the time constraints of this work, this implementation did not make it into the final solution.

4.3 Data Preprocessing

DFG GERiT. The data processing from DFG GERiT was completed with the aim of enhancing the handling of the data and making it easy to integrate into a database in a future phase. As the data was originally in JSON format, a vital step in preprocessing was the deserialization of JSON into C# objects. This transformation significantly facilitated more accessible and user-friendly interaction with the data within our C# codebase. Converting JSON into C# objects made the data more easily accessible, facilitating more effective manipulation using .NET's features.

Website Scraping. A vital preprocessing step was employed in the web scraping component to facilitate the natural language processing. It involved the conversion and extraction of HTML code into structured text, ideal for the analysis with NLP. For this, we used Textify, a package that specializes in transforming HTML content into a text-based format while discarding irrelevant or minimally informative elements. This involves deleting links to scripts, CSS files, and extraneous HTML syntax, along with various visual elements. The reasoning for this reduction closely relates to the limitations of present-day Large Language Models (LLMs). While these models are capable of reading and parsing HTML, their input size is restricted. By extracting only, the essential textual elements from the HTML content, we can improve the efficiency in processing the data.

To determine the most effective method for converting HTML to text, we conducted manual tests comparing SmartReader and Textify. We used university homepages as our test subjects and found that Textify outperformed SmartReader by converting entire pages. SmartReader primarily focuses on identifying articles within a page, which does not work well with most university homepages. An example of the output generated by Textify, illustrating the conversion from HTML to text, is presented in Listing 4.1.

Academics.com. For the data collected from Academics.com, we employed a preprocessing method resembling that of DFG GERiT. We deserialized the JSON data into C# objects to simplify its integration and manipulation in later stages of our

implementation. This procedure enhanced the data's manageability within our C# codebase, resulting in more seamless processing and analysis during the subsequent phases.

```
[IMG: Logo] [1]

++++++++++
Site title
++++++++++

Lorem ipsum dolor sit amet, consectetur adipiscing elit…

This is a table:
| Key | Value |
| One | Value |

[1] http://example.com
```

Listing 4.1: Example of converted HTML

4.4 Natural Language Processing

The Natural Language Processing (NLP) component of our solution focuses on using Large Language Models (LLMs) for structuring and processing unstructured data. The main goal is to use a web scraper to parse the rendered HTML source code of websites and further process this HTML into plain text. This text is then fed into an LLM, such as ChatGPT, to estimate the age of the professors from information available in their CVs or publication lists or to use the preprocessed data to find and structure job listings on that page.

For the implementation of our prototype, this process was streamlined to accommodate for the limited timeframe of this thesis. A C# implementation was developed using our Web Scraper API to retrieve the rendered HTML of manually entered website URLs, typically the profile page of professors or the university job portals. The HTML to text conversion, facilitated by Textify[22], significantly reduces the size of the data before it is processed by the LLM. This approach minimizes manual work and increases the efficiency of our NLP processing. Our NLP component integrates the Microsoft Tokenizer[23] to manage input size for the LLM. The tokenizer is set to limit the input context size to 10,000 tokens, aligning with most LLM processing capabilities.

[22] https://github.com/matteocontrini/Textify
[23] https://github.com/microsoft/Tokenizer

The OpenRouter[24] API facilitates the interaction with LLMs, providing a unified interface for querying different models. Inspired by the OpenAI API, the platform currently supports 43 models, including three beta models that are available for free for a limited time. The major models featured on the platform include OpenAI's GPT, Meta's Llama, and Claude. The decision to select OpenRouter was influenced by its ability to access a diverse range of models, which facilitates future model adoption and the availability of appropriate beta models at no cost that meet our application requirements. [39]

Moreover, by utilizing an API such as OpenRouter instead of a web-based implementation, like ChatGPT, we can provide system-level instructions to the models. This crucial feature guides the models' behavior with customized instructions and ensures more precise and tailored responses in line with our specific objectives. [58], [59]

Model Selection. During the development of our NLP component, we manually tested 10 different LLMs (see Table 4.1) to determine the best model for parsing text and generating JSON formatted responses. These tests included models trained for conversational tasks (Chat), general tasks, and models specialized for following instructions. Our selection criteria were based on the models' ability to accurately interpret and process the given text, with a focus on models that could effectively follow specific instructions and output structured data in JSON.

The decision to use a JSON output format was driven by its common use in LLM applications and the positive results it produced in our manual testing. As a lightweight and easily parse able format, JSON fits well with the data processing needs of our solution. Our investigation revealed that models designed for instruction-following tasks generally performed best for our requirements. However, it was the general-purpose models from OpenAI, specifically GPT-3.5 and GPT-4, that showed superior performance compared to most of the other models tested. Given the cost considerations, with GPT-4 being 3 to 10 times more expensive than GPT-3.5 depending on the specific variant, we decided to continue with GPT-3.5 for our implementation. This decision balanced cost efficiency with the high performance of the model.

Following our initial implementation, a new variant of the model, GPT-3.5 Turbo (1106), was released in November 2023. This updated model has a specialized JSON mode that perfectly suits our needs by forcing the model to output responses in a JSON-formatted string. After evaluating this new model, we found that it provided better results than the

[24] https://openrouter.ai/

older version of GPT-3.5 Turbo. As a result, we decided to integrate this latest model into our solution, taking advantage of its enhanced capabilities and specific JSON output feature. [60]

Model	Primary Application
toppy-m-7b	General
llama-2-13b-chat	Conversation
mistral-7b-instruct	Instruction
openchat-7b	Conversation
gpt-3.5-turbo-instruct	Instruction
gpt-3.5-turbo-1106	General
gpt-3.5-turbo	General
gpt-4-1106-preview	General
codellama-34b-instruct	Instruction
palm-2-chat-bison	Conversation

Table 4.1: Manually tested LLMs for our extraction Task

4.4.1 Vacancy Estimation

During the implementation of the vacancy estimation module, a crucial aspect was the creation of a customized prompt that instructs the LLM. This prompt, which is shown in detail in Listing A.1, was designed to guide the LLM through a four-step process: 1) extraction of relevant information from unstructured data, 2) estimation of the professor's age, 3) determination of success, and 4) formatting the age estimation output in JSON format. We adhered to our concept (see 3.5) and used a system-level prompt to assign greater weight to the prompt than to the actual input data, which was the HTML content that was converted to text. The whole estimation process is illustrated in Figure 4.3.

To validate our vacancy estimation process, we tested the profile links of professors from the Faculty of Computer Science at the University of Technology Chemnitz. These tests were particularly insightful as they provided real-world data for our model to handle. The model estimated the potential vacancy year of 11 professorships within the Faculty of Computer Science based on the estimated age of professors, whose profile pages were available. The calculation of vacancy year for each professorship was performed using

equations developed in the conceptual phase (see 3.5). This allowed us to predict the projected vacancy solely by utilizing the estimated ages of the current occupants.

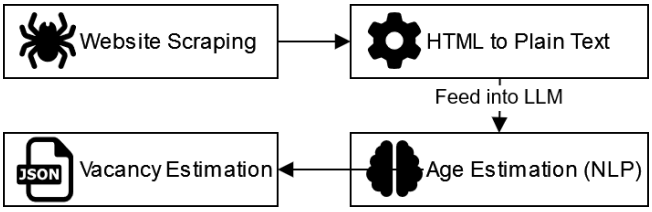

Figure 4.3: Vacancy Estimation Process using NLP

4.4.2 Parsing Job Listings

In addition to the job vacancy estimation module, we have developed the job listing parsing module, which is an integral part of our system, in a similar way. The basis of this module is a custom NLP prompt which is described in detail in Listing A.2, in combination with the chosen LLM (see 4.4). This prompt was designed instruct the model to perform several critical tasks in sequence: First, parsing and extracting relevant data from unstructured data; second evaluating the specificity and relevance of job roles, and finally, formatting this parsed information into structured JSON output.

This approach follows concept, as explained in Chapter 3.5. In this case, the input data is derived from the HTML content of the website, which is then converted to text before processing. The entire parsing process is visualized in Figure 4.4, which provides an overview of the workflow from data extraction to JSON formatting.

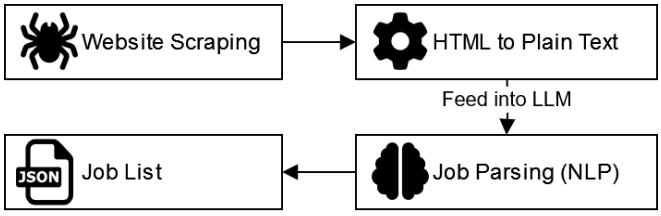

Figure 4.4: Job Parsing Process using NLP

4.5 Data Structuring

During the data structuring phase, our focus is on consolidating data from the diverse sources and NLP to establish a cohesive and uniform dataset. This is essential for the subsequent phase. For this purpose, we have chosen a NoSQL database approach

utilizing MongoDB for several reasons. MongoDB's schemaless design provides flexibility to accommodate diverse data structures from various sources, including structured information from databases and unstructured content from NLP processes. [61]

Our web scraping tools primarily utilize the .NET framework and are developed in C#. The compatibility between MongoDB client and .NET allows for seamless integration. Using the C# data models from our scraping and processing phases we streamline the data transition to the MongoDB [62]. MongoDB offers durable drivers and libraries for .NET, which simplifies database operations, such as CRUD (Create, Read, Update, Delete) actions, directly from our C# codebase [63]. To facilitate data presentation in the subsequent chapter, we stored the collected institutes from DFG GERiT, job postings from academics, and calculated ages from NLP in separate collections. We utilized the free tier of MongoDB Atlas for storage, enabling easy integration into the web application for presenting the data.

4.6 Data Presentation

In this section, we explore the presentation of data in the *ProfJobs* platform. This includes a detailed look at the backend and frontend architectures, the functionalities offered, and the hosting solutions used for the implemented web application. Our objective is to clarify the technical decisions made and how they contribute to the overall user experience and efficiency of *ProfJobs*.

Backend. For the web application's backend, we selected ASP.NET Core due to its compatibility with the .NET framework and C#, which are employed in other sections of our implementation as well. This enables us to capitalize on synergies, streamlining the reuse of data and previously established model objects within our web application. ASP.NET Core is a widely endorsed web framework, known for its openness and dependability. Over 3700 companies have contributed to its open-source repository. [64]

Its cross-platform design facilitates its implementation and functioning on different systems such as Windows, Linux, macOS, and Docker. We employed data adapters for each collection to manage data retrieval from our MongoDB database. Given the large sizes of the job and institution lists and the I/O limitations imposed by the MongoDB free tier, we have implemented memory caching and lazy loading techniques for these collections to optimize performance.

Frontend. The ProfJobs frontend is built using Blazor, a product in Microsoft's .NET ecosystem. Blazor is a framework for building interactive client-side web user interfaces with .NET, enabling us to share logic between server-side and client-side features. To expedite UI development, we leveraged the MudBlazor[25] extended UI library, which grants access to a diverse set of pre-built UI components. The user interface design adheres to the wireframe (see Figure 3.3) presented in the concept chapter, while the color scheme (see

Figure 3.2) is informed by the earlier analysis discussed in the concept chapter (see 3.7). The final product interface is displayed in Figure 4.5. [65]

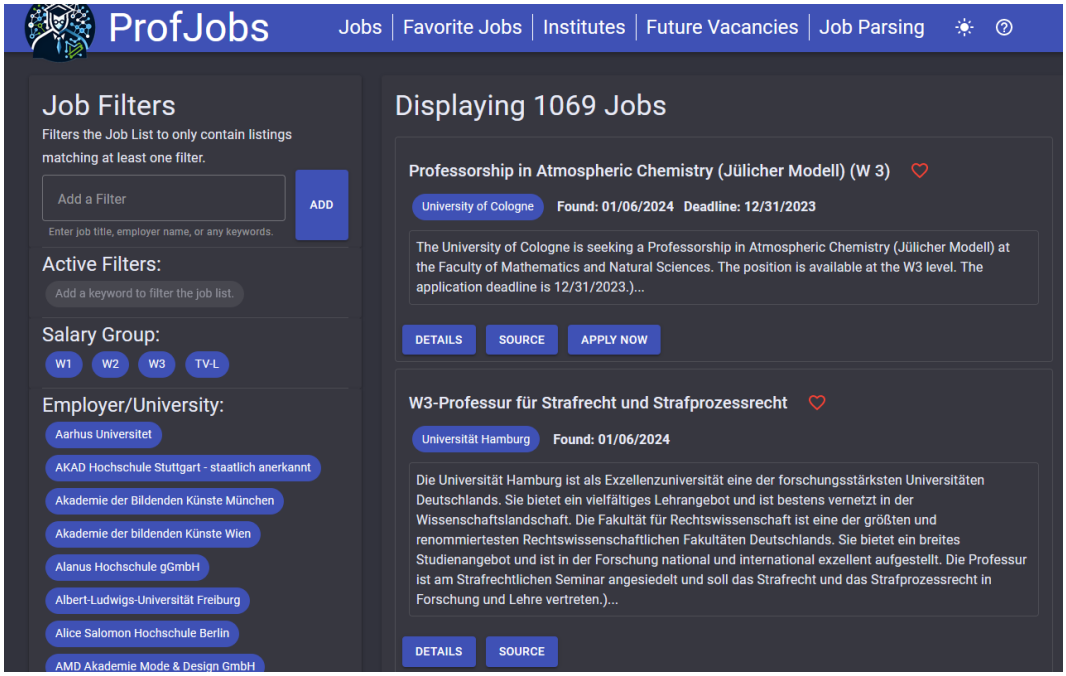

Figure 4.5: Screenshot of the ProfJobs Job List

Deployment. To streamline the deployment of *ProfJobs*, we created a custom Dockerfile that includes not only the application code but also all its dependencies. This Docker image then was subsequently deployed to a Linux server. This method significantly streamlines the deployment process and allows for quick and efficient updates, whether for introducing new features or fixing bugs. Our database is hosted on MongoDB Atlas[26],

[25] https://www.mudblazor.com/
[26] https://www.mongodb.com/atlas

ensuring an economical and productive hosting solution, while still being scalable for a potential future production phase of *ProfJobs*.

In addition to the main application, we have also taken important steps in deploying related components to improve the functionality of *ProfJobs*. Firstly, the Web Scraper API, which is an essential part of our data collection process, has been deployed as a Docker image on a Linux server as well. We also deployed the academics.com Web Scraper as a .NET tool on a Linux server. This tool was configured to run automatically once a day. Its main task is to collect new job listings and update existing ones. This automated process ensures that our database remains current and comprehensive, providing users with the latest job listings and insights.

Primary Functionalities.
Job List. The job list is a central feature, which provides an interactive and dynamic list of open professorship positions. It is designed to enhance the user experience by offering efficient job search capabilities. Users can easily search for job listings using any relevant keyword which is especially beneficial for users with specific requirements or interests.

Additionally, they can mark job listings as favorites to create a personalized list of preferred job listings as shown in Figure 4.5. Favorite jobs are added to a grid-based list, which is stored for the browser session. Users have the option to export selected or all favorite job listings in Excel or CSV format, making it easy to manage and share their job preferences. To assist new users in navigating these features, the job list section shows a tutorial on the first visit that guides users through three simple steps: finding a job listing, adding it to favorites, and exporting listings in Excel or CSV format.

Institutions. The institutions section, as shown in Figure 4.6, is a crucial component that provides users with comprehensive information about academic institutions. It includes a list of almost all institutions listed in the DFG GERiT database, offering users a broad overview of over 1800 academic institutions.

Users can search or filter through the institution list in a way like the job list. Each institution's entry includes a short description and additional information, such as the number of faculties, students, and professors. Additionally, direct links to the institution's job portal, primary homepage, and the homepages of their faculties, departments, and professorships are provided for easy access. If there are current job listings available at an institution, they are also displayed and highlighted, offering users a comprehensive

view of opportunities at each institution. Institutions can be searched by text or filtered by type, federal state, or custom keywords, like the job list.

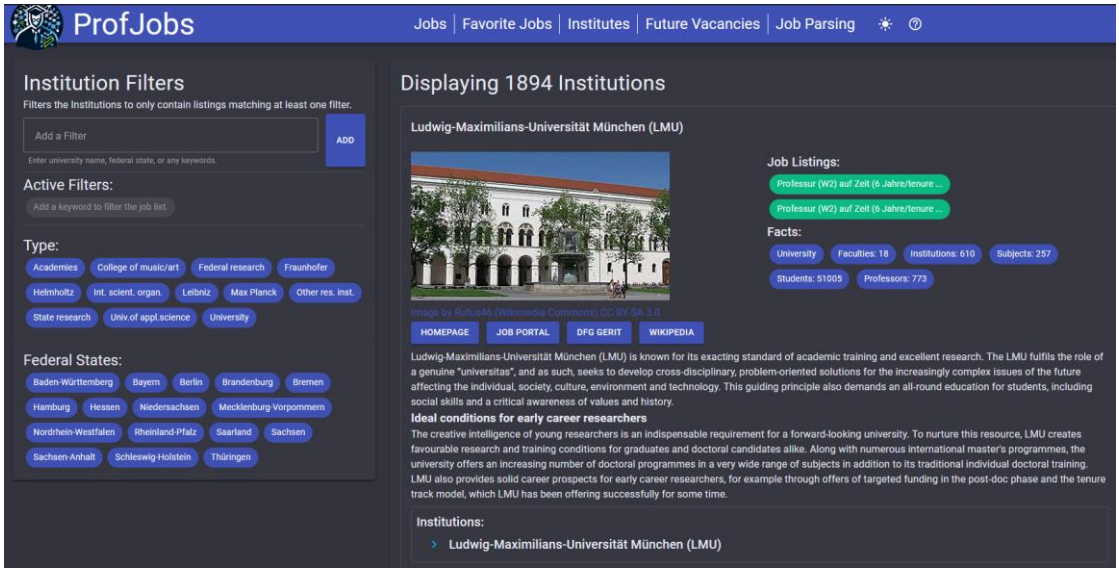

Figure 4.6: Screenshot of the ProfJobs Institution List

Future Vacancies. The future vacancies section enables users to estimate the potential future vacancy of a professorship by inputting the link to a professor's CV or profile page as shown in Figure 4.7. The page is automatically scraped, and the HTML content is converted to text and analyzed by an LLM to estimate and display the future vacancy. This section also displays previously estimated vacancies, including the name of the professorship and a URL to their website. To relate these estimated vacancies to a specific professorship, users can specify the GERiT identifier of that professorship, and the estimation will be saved.

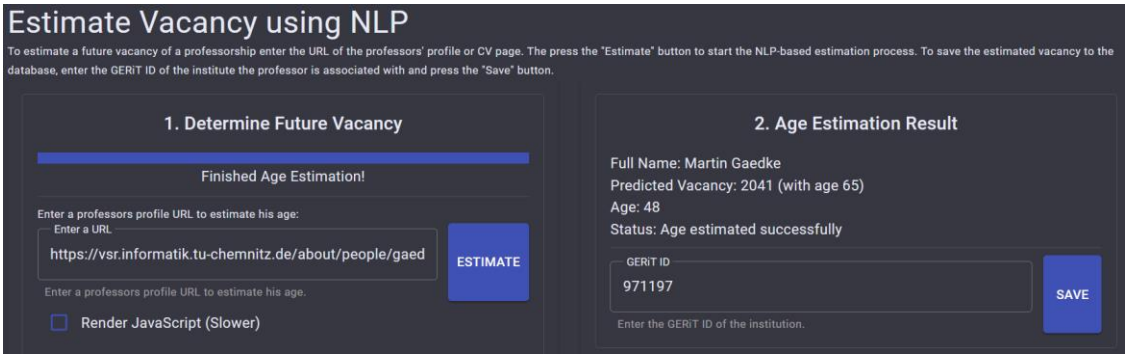

Figure 4.7: Screenshot of the ProfJobs Vacancy Estimation

Job Parsing. The job parsing section, as shown in Figure 4.8, offers a unique functionality that allows users to enter a URL, typically from a university's job portal. The URL will be scraped, the HTML content converted to text and then processed by an LLM to display the found jobs and its metadata in a structured format. The found jobs can be saved to the database for future reference. Previously parsed jobs are displayed on this page and included in the main job list, making them searchable alongside other listings.

Figure 4.8: Screenshot of the ProfJobs Job Parsing Section

4.7 Summary

This chapter covered the practical implementation of *ProfJobs*, our solution created to improve the academic job search process. It describes the development process, the technological decisions, and the reasoning behind in each phase.

In Section 4.1, the main implementation phases were introduced, beginning with data collection using web scrapers in Section 4.2 and followed by data preprocessing to refine the gathered information into a usable format in Section 4.3, as visualized in a BPMN diagram. Section 4.4 discusses the crucial role of the Natural Language Processing (NLP) phase in extracting structured content from preprocessed data, especially in estimating potential vacancy years of professorships. Section 4.5 covers the data structuring phase, which consolidates data from diverse sources and NLP outputs into a cohesive and uniform database. Section 4.6 discusses the data presentation of the *ProfJobs* platform, including the backend and frontend architectures, functionalities, and hosting solutions.

The following chapter will evaluate the effectiveness and impact of *ProfJobs* in the academic job search market.

5 Evaluation

In this chapter we perform an examination of *ProfJobs*, the web application developed as a part of this thesis. This evaluation is structured into two distinct parts, first we assess our solution in a user study, which is structured into three studies, each designed to focus on a specific aspect of the application and its utility to end users, particularly those seeking professorial positions. The second part is an assessment of our solutions against the previously developed requirements catalog.

5.1 Experiments

For the evaluation of *ProfJobs* two experiments were conducted to examine its usability, and data quality in the context of academic job search. The first experiment was a user study to examine the degree of user-friendliness and intuitiveness of the application from an end-user perspective. The second experiment assesses the reliability and comprehensiveness of the job listing data provided by *ProfJobs* against a manually created ground truth.

Each experiment utilizes a combination of empirical methods, such as user interactions with a prototype, questionnaires, or systematic data comparison. This approach guarantees that the evaluation covers a wide range of aspects that are essential for the success and relevance of the application in the context of the academic job search. The results of these experiments will not only provide insight into the current state of our solution, but also offer a direction for future improvements and enhancements. Additionally, we have planned an efficiency comparison study (see A.5), but due to the limited number of qualified participants we couldn't conduct that study.

5.1.1 Goals

In the evaluation of our solution, we intend to conduct a comprehensive analysis through two distinct experiments, each focusing on a critical aspect of the applications usability and data accuracy.

Study 1: Usability Evaluation.
This study evaluates the usability of the *ProfJobs* web application, which is a key factor in determining how easily end-users can interact with and utilize it. The independent variable in this study is the *ProfJobs* web application itself, and we are assessing its design and functionality in relation to user interaction. The study's dependent variables are the scores obtained from two different evaluation metrics: the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ). A high SUS score *(SUS > 68)* indicates a user-friendly system, which is essential for a satisfactory user experience [66]. The UEQ scores will provide insights into end-users' perceptions of the application's attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty, which are critical components of the overall user experience. [67], [68]

H1A: *ProfJobs* does achieve an above-average System Usability Scale (SUS) score.
H1B: *ProfJobs* does achieve an above-average score on the User Experience Questionnaire (UEQ) in all 6 dimensions.

Study 2: Data Quality.
The purpose of this experiment is to assess the accuracy and completeness of job listing data provided by *ProfJobs*, in comparison to university job portals. The independent variable consists of two categories: the *ProfJobs* platform and university job portals. The dependent variable is the level of data accuracy and completeness, which is measured by the degree of similarity with a manually created ground truth of job listings found on university portals. Our objective is to evaluate whether the platform can consistently reproduce the quality of information found in primary sources for vacant professorships, the university job portals. We will focus on achieving a minimum of 80% compliance in terms of data accuracy and completeness.

H2: *ProfJobs* does achieve over 80% compliance in terms of data accuracy and completeness compared to the job offers on university portals.

5.1.2 Methodology

Study 1: Usability Evaluation.
Material. The main material for this study is our operational web application prototype *ProfJobs*, implemented in Chapter 4, designed to include many features critical to the user experience. To evaluate the usability, participants will first complete two comprehensible practical scenarios to interact with the web application. The first scenario is about interacting with the job search and export functionalities of *ProfJobs* (see Listing A.4). The second scenario is about interacting with the NLP-based vacancy estimation

functionality (see Listing A.5). After these scenarios the participants complete the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ) survey. The SUS and UEQ were picked for its extensive acceptance and usability measuring reliability across digital platforms. Additionally, a demographics questionnaire will be given to participants to collect background information, such as age, gender identity, academic field, and current occupation as shows in Figure A.6. [66], [67], [68], [69]

Procedure. After engaging with *ProfJobs* in Scenario 1 (see Listing A.4), and Scenario 2 (see Listing A.5) participants will evaluate their experience with the platform in this study. The evaluation will begin with a demographic's questionnaire, followed by the SUS and the UEQ questionnaire. Both scenarios and the questionnaires will be presented in a predetermined order to ensure consistency in participant responses.
Instrumentation. The demographics questionnaire, as shown in Figure A.6, consists of 5 single-choice questions where multiple options are presented in a drop-down box and one free text question for the age. All question-and-answer options are listed in A.4. The study also employed a two-sided slider with 5 steps to administer the SUS, as depicted in Figure 5.1. For the UEQ questionnaires, a two-sided slider with 7 steps was used, as shown in Figure 5.2. The two-sided slider was preferred over the Likert-scale to better illustrate the linearity between the lower and upper boundaries.

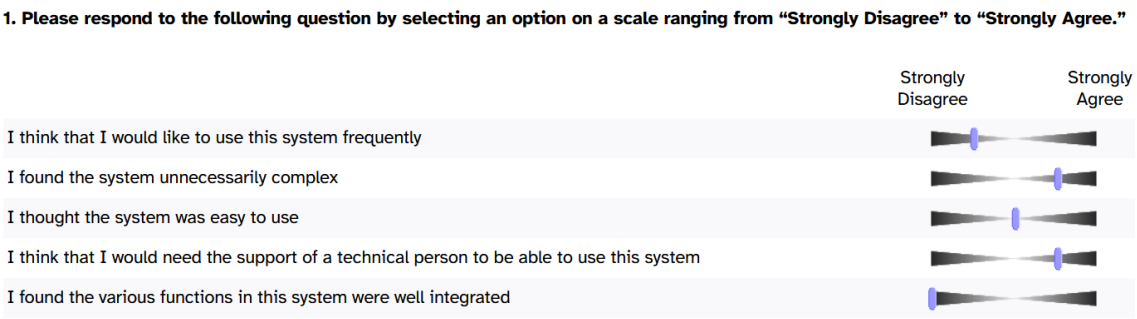

Figure 5.1: Part of our SUS questionnaire

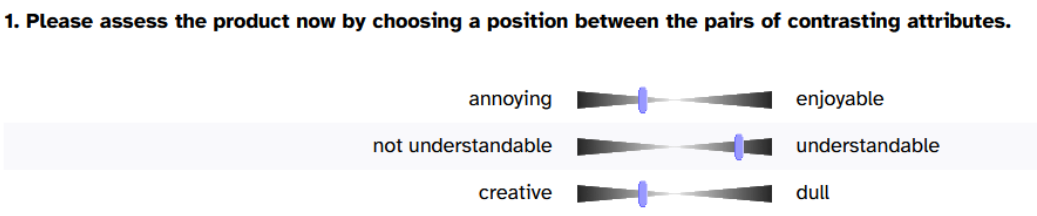

Figure 5.2: Part of our UEQ Questionnaire

Study 2: Results Quality.

Material. The study's primary material is a manually created ground truth dataset of job listings sourced directly from university job portals. For the creation of the ground truth dataset, we considered a total of 11 universities located in Germany (see Table A.2). The first 10 universities were selected using the DFG GERiT database by the largest number of students, as addition we also added the Chemnitz University of Technology as the 11th university.

The following meta information were manually collected for each job listing: Job Title and Details, Salary Group, Employer, Application Deadline, Date Posted, Apply URL, Apply Email. In total we obtained 36 job listings from these universities. If there was an English version of the job listing available, it was favored. The data was collected on 29.12.2023. This ground truth dataset (see B.2) serves as a benchmark for evaluating the accuracy and completeness of job data provided by *ProfJobs*.

After the manual collection, *ProfJobs* was used on the same day to scrape all professorial job listings from Academics.com and the job portals of these universities were parsed using the NLP job parsing component of *ProfJobs*.

Procedure. To align the job listings from *ProfJobs* with those in the ground truth dataset, unique identifiers for the job listings were introduced and the job listings were manually matched to the ground truth. This step is necessary to compare each pair of listings (*ProfJobs* vs. ground truth) to evaluate the similarity and completeness of the job listings in relation to the ground truth. When there were multiple job listings in *ProfJobs* for the same job the one sourced from the NLP component was preferred.

Instrumentation. For measuring the similarities between the ground truth and the data from *ProfJobs* we use the Levenshtein distance. The Levenshtein distance, often denoted as lev(i,j) (5.1), is a metric that quantifies the number of single-character edits, such as insertions, deletions, or substitutions, needed to transform one word *i* into another word j. The similarity score is calculated using the Equation (5.2). This formula gives a higher score to strings that are more similar, with 100 indicating an exact match. This metric was chosen because it captures the text variations that are critical to our analysis of job postings. [70], [71]

A custom Python implementation (see B.2) was used to calculate the Levenshtein distance and similarity for each job offer. In addition, the average Levenshtein distance and similarity were calculated for each individual job listings and for all job listings.

$$\mathrm{lev}_{a,b}(i,j) = \begin{cases} \max(i,j) & \text{if } \min(i,j) = 0, \\ \min \begin{cases} \mathrm{lev}_{a,b}(i-1,j) + 1 \\ \mathrm{lev}_{a,b}(i,j-1) + 1 \\ \mathrm{lev}_{a,b}(i-1,j-1) + 1_{(a_i \neq b_j)} \end{cases} & \text{otherwise.} \end{cases} \tag{5.1}$$

$$\text{Similarity} = \left(1 - \left(\frac{\text{Levenshtein Distance}}{\text{Max Length of two strings}}\right)\right) \times 100 \tag{5.2}$$

5.1.3 Results

In this section, we present the objective findings of our studies, which focuses on the usability of *ProfJobs* for professors, postdocs, and senior PhD students. The whole usability study, including the demographics questionnaire was conducted by 19 participants in total. However, out of the 19 participants only 14 conducted the full survey and therefore were considered in the following parts of the evaluation. The participants of our user study were found via direct contacts, reddit and discord as well as special pages for sharing surveys. The study was conducted online with the design introduced in 5.1.2 using the SoSci survey platform[27], from 04.01.2024 to 21.01.2024. The results are presented as directly measured and observed that reflect the participants' responses without any subjective interpretation.

Demographics Questionnaire.
This section presents the demographic characteristics of the participants in our study. The data collected provides information on various aspects such as age, gender, level of education and current employment.

Age distribution: The age of the participants ranged from 26 to 43 years. The median age was 32.5, reflecting a central tendency towards a younger middle age.
Gender: The gender distribution, visualized in Figure 5.3, among participants was the same for female and male participants with 6 each. Two persons chose "Prefer not to say" instead of stating their gender. Notably, there were no participants who identified as non-binary/third gender or the other category.

[27] https://www.soscisurvey.de/

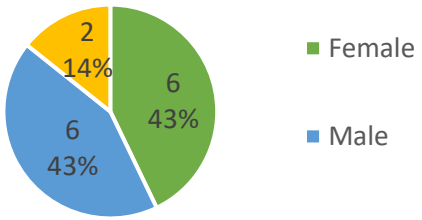

Figure 5.3: Demographics: Gender Distribution

Academic Background: The participants in this study are predominantly from the field of computer science with 8 out of 14 participants, reflecting a good fit of this subject to our scenario. In addition, various scientific disciplines are represented: one participant each from the fields of natural sciences, formal sciences, and medicine and health sciences, as well as two participants from the field of applied sciences.
Current Occupation: The vast majority, 7 out of 14 participants, are postdocs as visualized in Figure 5.4. This is complemented by the presence of two associate professors. In addition, 5 participants are currently doctoral students.

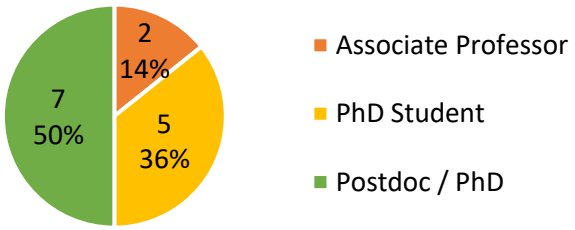

Figure 5.4: Demographics: Occupation Distribution

Mother Tongue and Translation Tool Usage: Most participants, 11 out of 14, stated that they did not use a translation tool. This non-use of translation tools extends to both non-German and German native speakers. Interestingly, the 3 participants who used a translation tool were both non-native German speakers. Overall, 8 out of 14 participants stated that their mother tongue is German.

Country currently based in: The participants are based in Germany (5), India (3), Bulgaria (2), Argentina (1), Russia (1), Greece (1) and Austria (1). Interestingly, most of our participants are based in the EU.

System Usability Scale.
The average SUS score over all questions measured in our study was 79.64. This score reflects the overall usability of *ProfJobs* as perceived by the participants. The average normalized SUS scores for each question are ranging from 5.36 for Question 5 to 9.46 for Question 10 as shown in Figure 5.5. The highest variation in responded scores was with Question 1.

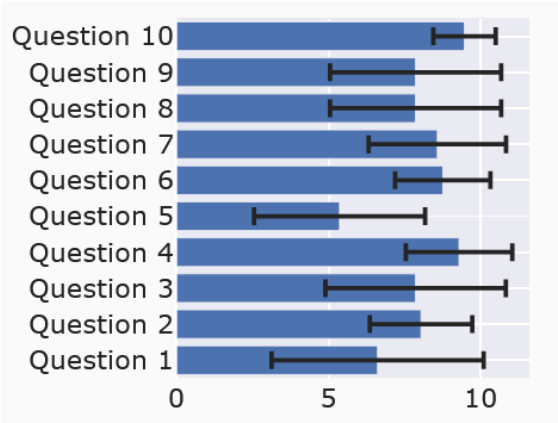

Figure 5.5: Normalized average SUS-Score per Item

User Experience Questionnaire.
The UEQ results are presented in 6 different dimensions, each of which provides valuable insights into different aspects of the usability of the system and the overall user interface. These dimensions include attractiveness, perspicuity, efficiency, dependability, stimulation, and novelty.

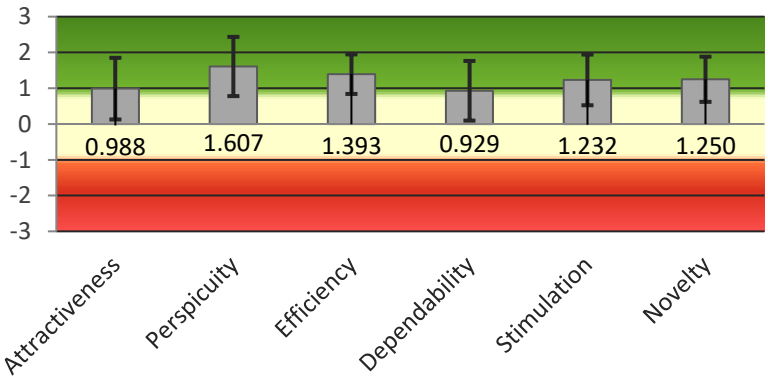

Figure 5.6: UEQ: Average scores visualized according to Blattgerste et al. [72]

In Figure 5.6 the mean across each dimension was visualized on a scale ranging from -3 (horribly bad) to +3 (extremely good) visualized with vertical bars. The measured mean

scales of our UEQ range from +0.929 to +1.607. The highest mean with +1.607 was recorded for perspicuity, which assesses the clarity and comprehensibility of the system. For dependability the lowest mean of +0.929 was recorded. For attractiveness a mean of +0.988 was recorded, which is a measure of the aesthetics of the system. In the efficiency dimension, which reflects the performance of the system, a mean of +1.393 was recorded. The stimulation dimension, which indicates the system's ability to engage and motivate users, was recorded with a mean of +1.232. Finally, in the novelty dimension, which assesses the innovation and freshness of the user experience, the systems mean of +1.250 was recorded. Notably the means across all dimensions are positive.

Study 2: Results Quality.

A total of 36 job listings were manually retrieved as ground truth from the selected university portals, which represents the dataset for the evaluation. In addition, we used *ProfJobs* to retrieve 27 job listings for professorships from both Academics.com and the university portals which were parsed using the NLP component. These job listings were then manually matched with our ground truth for further analysis.

The majority of job listings in the ground truth with an amount of 12 are from the Baden-Wuerttemberg Cooperative State University, followed by the Universität Hamburg with 5 job listings. The smallest number of job listings with only one is from the Chemnitz University of Technology. Most job listings from the exported *ProfJobs* data with an amount of 17 out of 27 were obtained using the NLP component. The other 10 job listings were obtained by scraping component of *ProfJobs* on Academics.com.

In Figure A.9 the Levenshtein distance and in Figure A.10 the similarity matrix were visualized in an heatmap ranging from green signaling a high similarity or low distance to red signaling a low similarity or low distance. The distance comparison shows an average distance of 20.32 across all job listings. The average similarity across all job listings measured was 63.03%.

5.1.4 Discussion

Study 1: Usability Evaluation.

System Usability Scale.
In our study, we hypothesized that *ProfJobs* would achieve an above-average score on the SUS, indicating good usability. For the analysis of the SUS, we have utilized the widely used SUS Analysis Toolkit[28] by Blattgerste et al. [72].

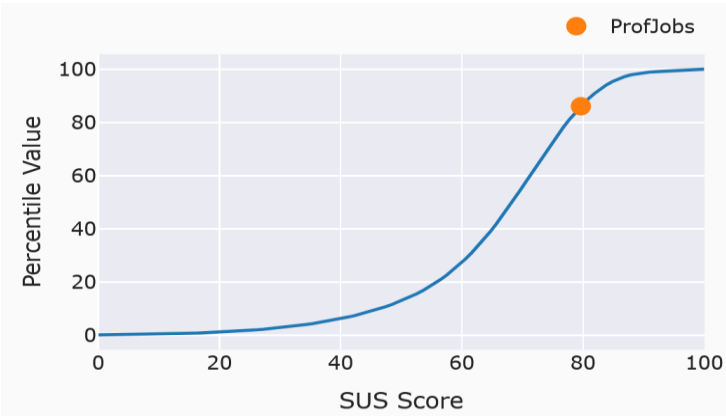

Figure 5.7: Visualization of SUS on cumulative percentile curve of Sauro et al. [73]

Our SUS evaluation resulted in an average perceived **SUS of 79.64**, calculated based on the formula from Brooke et al. [69]. Compared to the industry average of 68, *ProfJobs* has achieved an above average score with an adjective scale of "Good". These results indicate the support of our hypothesis H1A, that *ProfJobs* has an above average user experience on the SUS. In Figure 5.7 we plotted our SUS on the cumulative percentile curve from Sauro et al., which was derived from mapping over 5000 SUS scores [73]. This visualization allowed us to position the usability of *ProfJobs* in the context of a wide range of systems. Remarkably, our score places us in the 86th percentile, meaning that the usability is higher compared to 86% of the systems in the Sauro et al. dataset [73].

Based on the number of participants in our study, 14 in total, the conclusiveness diagram of Tullis et al. suggests that our SUS has a degree of conclusiveness of 100% as shown in Figure 5.8. This indicates a high reliability of our results and provides confidence in our findings, which further supports our hypothesis. [74]

[28] https://sus.mixality.de/

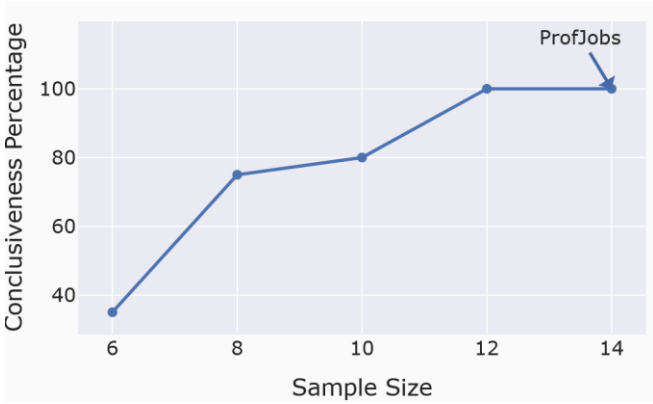

Figure 5.8: Conclusiveness Percentage based on Tullis et al. [74]

Figure 5.5 visualized the normalized items of the SUS representing their contribution to the SUS score. These show that certain aspects of *ProfJobs*, particularly Question 10, were rated highly which indicates that the interface is rather easy to use and does not require to learn a lot of things before using. Conversely, Question 5 may require additional attention and improvement due to its low score, which indicated that the functionalities are not well integrated within *ProfJobs*. These findings are crucial for targeted improvements in future iterations of the platform and to further improve the overall SUS score.

Figure 5.9 shows a boxplot representation of the results of the SUS for *ProfJobs*. This graphical representation highlights the distribution and spread of the usability scores and provides a more differentiated understanding of the user perceptions. The average SUS score for *ProfJobs* is 79.64, which significantly exceeding the industry standard of 68, which puts the platform in the "Good" category on the adjective scale and an "A" on the grading scale. Such a high mean score indicates a generally positive user experience.

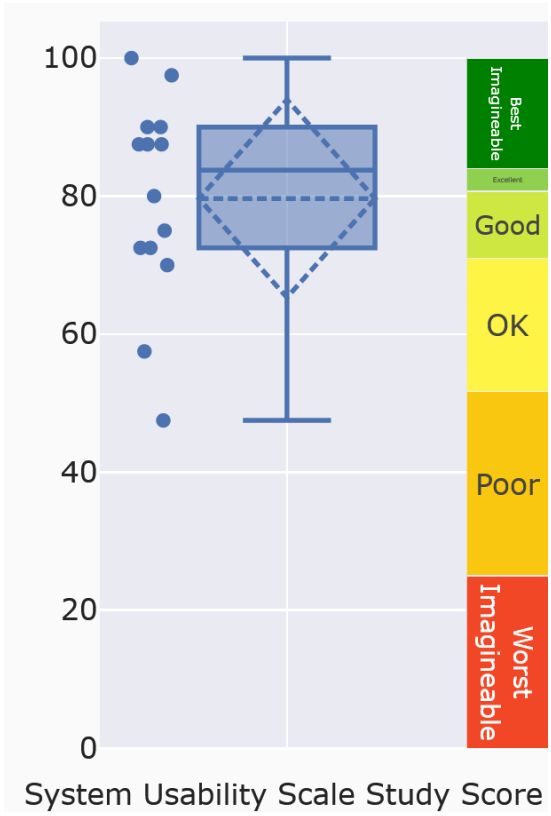

Figure 5.9: SUS: Average perceived usability score

The boxplot also shows a standard deviation of 14.33, which indicates a moderate spread of user opinions. This is evidenced by the range of ratings from a minimum of 47.5 to a maximum of 100, which indicates that although many users consider the system to be very useful, there are also outliers with less positive opinions. The interquartile range, from 71.88 to 90, shows that the middle 50% of ratings are at the upper end of the scale, with a median of 83.75, which is even higher than the mean, indicating a positive skew in the data.

This boxplot in Figure 5.9 is an important addition to the previous figures as it provides a visual and statistical perspective on users' overall satisfaction with *ProfJobs* and further supports our hypothesis H1A. It supports our assertion that *ProfJobs* provides an above average user experience compared to many other systems in the industry, as evidenced by its positioning in the 86th percentile.

The results of our SUS evaluation indicate that *ProfJobs* meets the criteria for an above average usability rating. This result underscores the effectiveness of *ProfJobs'* user interface and overall ease of use. To further evaluate the statistical significance of these findings, we conducted a one-sample t-test [75] with our measured mean SUS of 79.64 and a standard deviation of 14.33, against the industry average of 68 according to Sauro et al. [76].

The resulting positive t-statistic of 3.03 indicating that our mean SUS is significantly higher than the industry average and outside the expected range of the null hypothesis. Given the p-value of 0.0047, which is above the strictly defined significance level of $\alpha > 0.005$ according to Benjamin et al. [77], we find statistical evidence to reject the null hypothesis. These results further support our hypothesis H1A, that ProfJobs has an above average user experience and that our findings have significant evidence.

User Experience Questionnaire.
Our hypothesis H1B refers to the fact that ProfJobs performs above average in the 6 dimensions of UEQ. For the analysis we have utilized the UEQ Data Analysis Tools Version 12[29] by Schrepp et al. [78].

[29] https://www.ueq-online.org/

During the analyzation of the data, we found inconsistencies in the participants' responses, which necessitated a data integrity check. Of the 14 responses collected, 2 were identified and removed as they contained 3 critical inconsistencies. We applied two heuristics according to the UEQ manual guidelines [78]: First, we identified and removed responses where the variation within a scale exceeded a difference of 3, indicating non-serious responses or misunderstandings. Second, we excluded responses with a consistent pattern, especially when more than 15 items received the same rating, indicating a lack of engagement.

We compared *ProfJobs* average scores on the UEQ scales with a benchmark dataset comprising data from 21,175 participants from 468 studies which is updated once every year. For our analysis we have utilized the latest version 12 as of the time writing. This dataset covers a range of products, from enterprise software to social networks, and provides a comprehensive context for our assessment. [79]

The UEQ benchmark for *ProfJobs*, as shown in Figure 5.10, shows varying performance across the different dimensions of user experience. In particular, the product performed below average in the areas of attractiveness (m=1.08) and dependability (m=0.79), placing it in the bottom 50% of the benchmark. This suggests that there is a need for improvement in these areas. In particular, attractiveness could be improved by refining the visual design and user interface, while dependability could benefit from focused efforts on predictability and security of user interactions, so that the user feels more in control.

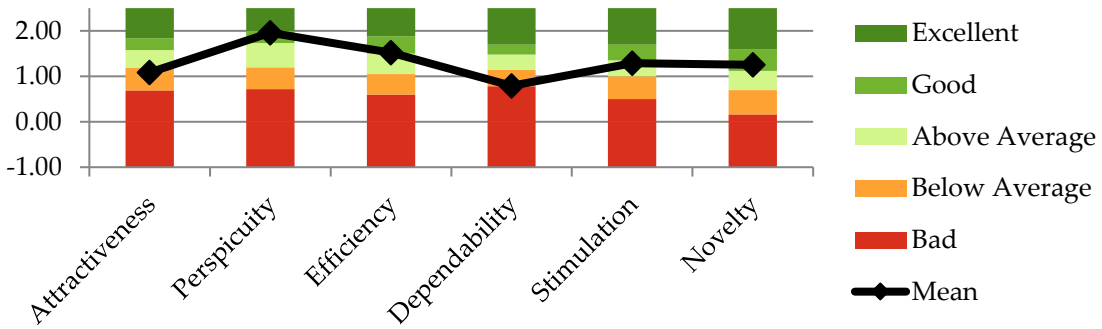

Figure 5.10: ProfJobs UEQ Benchmark against 468 studies

On a positive note, *ProfJobs* scores good in terms of perspicuity (m=1.96), which indicates that users generally find the product easy to understand and clear. This is a significant strength, as clarity of use is an important component of user satisfaction. However, there

is still room for further improvement to reach the top quartile in this dimension, possibly through iterations of the platform based on user feedback from the user group.

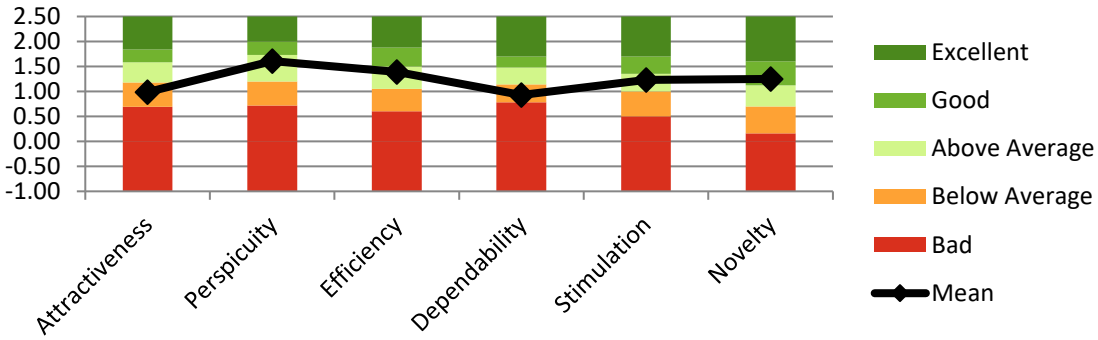

Figure 5.11: ProfJobs UEQ Benchmark against 468 studies with critical data points

ProfJobs also achieved a good score for efficiency (m=1.96) and an above average in stimulation (m=1.29). This suggests that it is perceived as more efficient at facilitating tasks and stimulating users for interactions, compared to other products. To further improve the user experience in these dimensions, optimizing task flows for more efficiency, incorporating more engaging content or interactive features for more stimulation could be added.

Another good rating was achieved in the novelty (m=1.25) dimension, suggesting that the functionalities of *ProfJobs* are unique and creative, catching the users interest. However, compared to other systems there is still room for improvement, which could be achieved with a more creative design of the user interface.

Overall, the results are positive but also underline that there is a need for further improvements and development. Maintaining strengths in novelty, perspicuity, efficiency, and stimulation while actively working on identified areas of improvement such as attractiveness and dependability will be key to improving the overall user experience and competitive position. Interestingly the benchmark of the UEQ with the critical datapoints, visualized in Figure 5.11, shows a lower performance across all dimensions but dependability compared to Figure 5.10 where these data points were removedFigure 5.8.

In our user study on *ProfJobs*, the standard deviations shown in Table A.1 for the various UEQ dimensions are higher than the suggested thresholds in Schrepp et al. studies [80]. These thresholds, which are based on an analysis of numerous UEQ studies, categorize the standard deviations as an indicator of high, medium, or low agreement:

- **High Agreement:** Standard deviation below 0.83
- **Medium Agreement:** Standard deviation between 0.83 and 1.01

- **Low Agreement:** Standard deviation above 1.01

As all standard deviations in your study are above 1.01, this indicates a low level of agreement among the participants. This could be interpreted as a sign that different users had different experiences or perceptions of the UX aspects measured by the UEQ. The relatively small sample size (N=14-2) in your study may also contribute to the high standard deviations. A small sample size may lead to less reliable estimates of population parameters such as mean and standard deviation. Therefore, care should be taken when interpreting our results as they may not reflect the opinion of a wider user base. Increasing the sample size in future studies could help to achieve more stable and generalizable results.

In view of hypothesis H1B, which states that *ProfJobs* performs above average in all six dimensions of the User Experience Questionnaire (UEQ), the results of the one-sample t-tests [81] provide a differentiated picture. For all 6 dimensions, the conducted t-tests did not provide sufficient evidence to conclude that *ProfJobs* is above the industry average. The p-values in these categories were significantly higher than the suggested threshold of $\alpha > 0.005$ according to Benjamin et al. [77] for statistical significance, which led to the null hypothesis being retained.

In conclusions the benchmark results partially support H1B. While *ProfJobs* excels in perspicuity, efficiency, stimulation, and novelty above the industry standards, we could not find statistical evidence using the one-sample t-test for our results. This highlights certain areas where *ProfJobs* meets user expectations, but also points to areas for improvement, particularly in terms of making the platform more attractive and reliable (dependability). Given these mixed results and the small sample size, further research with a larger group is recommended to confirm these findings and gain a more comprehensive understanding of the user experience on *ProfJobs*.

Study 2: Data Quality.

Our analysis aimed to evaluate the compliance of *ProfJobs* in terms of data accuracy and completeness compared to job listings on university portals. The hypothesis to be tested, H2, states that *ProfJobs* achieves over 80% compliance in this regard. To assess the validity of this claim, we conducted a one-sample t-test by calculating the similarity values between the ProfJobs data and ground truth from the university job portals.

The results of our analysis revealed an average similarity of around 63.03% (see Figure A.10). This value is significantly below the threshold value of 80% postulated in our hypothesis. Further statistical analysis using the t-test resulted in a negative t-statistic of -6.36 and an extremely small p-value of 8.22×10^{-7}, which is approximately zero. The negative t-statistic indicates that the mean value of the sample is significantly below the hypothesized benchmark value of 80%.

Given the small p-value, which is well below the refined significance level ($\alpha > 0.005$) [77], we can support the null hypothesis. This means that there is substantial statistical evidence that the average similarity value is not at least 80%. In other words, the data accuracy of *ProfJobs* compared to the job listings of the university portals is significantly below the assumed level of similarity.

However, it is important to consider these results in context. Although *ProfJobs* did not reach the assumed threshold of 80%, the system performed remarkably well. The system was able to successfully retrieve 27 out of 37 job offers on the university portals, which equates to a retrieval rate of around 75%. This efficiency is highly creditable for an automated system, especially considering the complexity and variability of analyzing and extracting job listings from different unstructured sources using NLP. Also, the measured Levenshtein distance is considered low with an average of only 20.32 as shown in Figure A.9. While this aspect of *ProfJobs* performance is not consistent with the original hypothesis, it nonetheless highlights its practical utility and effectiveness in a real-world application context.

5.2 Requirements based Evaluation

In this section, we analyze our implemented solution, *ProfJobs*, against the 6 predefined requirements (see 2.1). Our aim is to evaluate its performance and utility for postdocs searching for a professorship. By assessing the extent to which *ProfJobs* meets each requirement, we provide a clear evaluation.

R1: Low-Effort Rating: ★★★★ (Good)
ProfJobs integrates various automation features that drastically minimize the need for manual data entry. Once the job filters are set up, there is no need for recurring manual intervention since it automatically filters all job listings or institutions according to the users' needs. Furthermore, new job listings are automatically collected daily without any user input. The advanced search features and the ability to export selected job listings

into standard tabular formats such as Excel and CSV further enhance the low-effort nature of our solution. This is also supported by the conducted user study in 5.1.4. Based on these factors, *ProfJobs* is rated "good" in fulfilling this requirement.

R2: Ease-of-Use Rating: ★★★☆ (Fair)

- **Search functionality:** A full search feature and filters are offered.
- **Accessibility:** The adherence to web standards partially ensures accessibility.
- **Export functionality:** Jobs can be exported in tabular formats like Excel or CSV.
- **Documentation:** A 3-step introduction tutorial is shown on the first visit.
- **Recommendations:** A recommendation feature is not available.
- **Multi-device support:** Desktop and Mobile devices are supported.

Given that it fulfills most of the defined ease-of-use criteria and as its supported by the conducted surveys in 5.1.4, *ProfJobs* is rated as "fair" in this category.

R3: Versatile Data Integration Rating: ★★★★ (Good)
The solution incorporates data from various sources such as DFG GERiT, university websites, and job portals like academics.com. These sources are publicly available, which enhances the versatility of data integration. As a result, *ProfJobs* receives a "good" rating for its versatile data integration.

R4: Completeness of the Data Rating: ★★★☆ (Fair)
ProfJobs offers an extensive database of academic institutions and job postings, providing information on aspects such as country, state, university, institutional type, website URLs, deadline dates, and salary groups. It also allows for the estimation of future vacancy dates. Nevertheless, certain data points, like the current occupant of the chair or professorship, are solely present in the job descriptions' thorough textual content or missing entirely. Considering the availability of most data points, *ProfJobs* is rated as "fair" in terms of Completeness of the Data.

R5: Data Refresh and User Preference Retention Rating: ★★★★ (Good)
Users can store their preferences, such as favorite job listings for their browser session. The job listings refresh automatically once per day via the academics.com data source. These functionalities meet this requirement, resulting in a "good" rating for Data Refresh and User Preference Retention.

R6: PhD-Centric Solution Rating: ★★★★ (Good)
ProfJobs has been specially designed to meet the needs of PhDs looking for vacant professorships. It provides distinctive job search and filtering features customized for this

group, including the ability to predict potential future openings via NLP-based estimations as well as parsing job listings using NLP. This emphasis on tracking long-term opportunities for individuals with PhDs justifies rating *ProfJobs* with "good" for meeting the requirement.

Summary.
This chapter assessed the strengths and weaknesses of our solution *ProfJobs*. According to our evaluation *ProfJobs* is an outstanding solution for the academic job search, which is characterized by high performance in several requirements. It excels in Low-Effort (R1), Versatile Data Integration (R3), Data Completeness (R4), Data Refresh and User Preference Retention (R5), as well as being a PhD-Centric Solution (R6).

Our solution effectively bridges the gap between detailed information retrieval and user-focused features by utilizing web scraping to ensure comprehensive data, all while maintaining an intuitive and low-effort experience for users, which is also supported by the conducted user study in 5.1.4.

Req. / **Approach**	**Job Portals**	**University Homepages**	**NLP-Based**	**Web Scraping**	**Search Engine**	**APIs**	**ProfJobs**
R1	★★★★	★☆☆☆	★★☆☆	★★☆☆	★★★☆	★★☆☆	★★★★
R2	★★★★	★★☆☆	★★☆☆	★☆☆☆	★★★☆	★☆☆☆	★★★☆
R3	N/A*	★★★★	N/A*	★★★★	★★☆☆	★★★★	★★★★
R4	★★★★	★★★★	N/A*	★★★★	★★★☆	★★★★	★★★☆
R5	★★★★	★★★☆	★★★★	★★★☆	★★★★	★★★☆	★★★★
R6	★★☆☆	★★★☆	★★☆☆	★★☆☆	★★☆☆	★★★☆	★★★★
∑	**18**	**17**	**10**	**16**	**17**	**17**	**22**

Table 5.1: Evaluation matrix for our solution

In summary, *ProfJobs* showed significant advance among the evaluated job search solutions with the highest score of 22. It offers a unique and powerful tool for professors, postdocs and senior PhD students seeking a professorship, by effectively combining the depth of information from academic sources with the user-centric features of modern job portals. It addresses the specific needs of this demographic in a way that other approaches in our investigation do not, making it an invaluable resource and addition for the current academic job search market.

5.3 Threats to Validity

This section addresses potential threats to the validity of the conducted experiments. The evaluation of our test setup is based on the main validity threats: internal, external, and construct validity, each of which addresses different aspects of the study's design and implementation. Recognizing and addressing these threats is critical to ensure the credibility of our experiments and providing a full understanding of the results limitations. [82], [83], [84]

Internal validity.
Internal validity refers to the degree to which a research study can establish a causal relationship between two variables and exclude other possible explanations. To achieve internal validity, a study should fulfill three criteria: temporal precedence, covariation, and no plausible alternative explanations. [82]

The usability assessment benefits from the use of established assessment tools such as SUS [69] and UEQ [67], but participants' prior experience could also influence perceptions, potentially affecting the study's ability to draw firm conclusions about the usability of the application. In the data accuracy study, direct comparison with a ground truth dataset provides a solid framework for assessing the accuracy and completeness of the resulting data. However, the manual method used to create this ground truth dataset and the selection criteria of the university portals could have introduced a potential bias.

External Validity.
External validity refers to the extent to which the results of a study can be generalized to other attitudes, population groups or situations [82]. The usability assessment attempts to improve generalizability through broad participation of professors, postdocs, and senior PhD student, but the results may not be transferable to users with a different technological background or outside the academic sector. For the data accuracy study, the generalizability of the results depends mainly on the representativeness of the baseline dataset. The dataset only contains job listing data from 11 German universities, which is why the results may not be generalizable for all universities or universities outside Germany.

Construct Validity.
Construct validity refers to the accuracy with which the study design measures and tests the intended concepts and variables [85]. The usability assessment relies on the SUS [69] and UEQ [67], which are robust instruments and give a good reference point for the

usability of *ProfJobs.* However, as these questionnaires are created for general systems, they might not map all relevant aspects to usability and user experience specific to *ProfJobs*. In the data accuracy study, Levenshtein distance is used as an objective measure.

5.4 Summary

In this chapter, we evaluated *ProfJobs* in a user study with 14 participants using the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ). The SUS results show a mean score of 79.64, which is significantly above the average and indicates a good level of user-experience. This result was further supported by a one-sample t-test with a t-statistic of 3.04 and p of 0.0047.

The UEQ results were mixed: ProfJobs scored highest in perspicuity, efficiency, and novelty with a good rating. While achieving an above average rating in stimulation and a below average rating in attractiveness and dependability. These results suggest that while users find the platform appealing in most dimension on the UEQ scale, its usability could still be improved in certain areas.

An important part of our study was the assessment of data quality, where we analyzed the accuracy of job listings from German universities. The results showed an average similarity of 63.03% and a retrieval rate of 75%. Although our target match rate of 80% was not achieved, these findings suggest that *ProfJobs* is quite effective in finding relevant vacancies, although data accuracy can still be improved.

We also compared ProfJobs with current solutions and found it to be superior in meeting specific user requirements for the user group. However, the evaluation showed that certain aspects of usability need to be improved compared to solutions of the state-of-the-art. Finally, we addressed threats to validity, acknowledging limitations such as the small sample size and the study's focus on a specific user group, which could affect the generalizability of the results.

In summary, while ProfJobs shows promise, particularly in terms of aesthetics and overall usability, the study also highlights several areas where usability and data accuracy can be improved.

6 Conclusion

This thesis began with an in-depth examination of today's academic job market, critically examining the prevailing challenges and shortcomings of existing methods for finding professorial positions, focusing on the perspective of postdocs. This initial discourse provided the foundational context for the subsequent analytical examination of current state-of-the-art solutions. A customized list of requirements was meticulously formulated and served as a benchmark for the evaluation of the various existing approaches. This investigation brought to light a striking gap in the current landscape: the methods examined are not able to fully meet the required criteria. This finding underlines the need for a more efficient and integrated solution.

In response to this gap, this thesis developed a novel solution concept for a platform which combines the most promising aspects of the evaluated state-of-the-art methods in a unique way. Our solution *ProfJobs*, developed with .NET 8 and a Blazor front-end, represents a significant step forward in the area of academic job search and is characterized by its multifunctional capabilities. Key features include a web scraper for Academics.com, an advanced NLP component to convert unstructured HTML into structured job listings, and a predictive NLP model for future professor vacancies that uses a Large Language Model (LLM) to estimate the age of professors based on their CV or online profiles.

In addition to its technical capabilities, *ProfJobs* is characterized by its user-centric design. It enables an intuitive and efficient search process for vacant professorships, which includes over 1000 job listings and detailed data on more than 1800 German institutions. with the system automatically updating and adding new listings daily. The user interface is specifically tailored to the needs of postdocs and senior PhD students, allowing them to easily navigate and filter the listings.

The benefits and user-friendliness of *ProfJobs* were empirically validated in a user study with a representative group of the target users consisting of professors, postdocs, and senior PhD students. The results were quantified using the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ). The SUS clearly confirmed *ProfJobs* as a user-friendly platform with an above average **SUS score of 79.64**. The statistical significance was further supported with a one-sample t-test. The UEQ supported that *ProfJobs* is **superior in the dimensions of perspicuity, efficiency, stimulation, and novelty** but

also suggests that improvements can still be achieved in the area's attractiveness, and dependability.

In addition, an assessment of data quality using the Levenshtein distance metric for a side-by-side analysis with a manually created ground truth of vacant professorships resulted in a **similarity index of 63.03%**. Although this score does not reach the target threshold of 80%, it is noteworthy that *ProfJobs* **independently found and parsed 75%** of the job listings from university portals which were included in the manually created ground truth, most of which were found using the NLP component. This demonstrates its real-world efficiency. When compared against the same criteria as the state-of-the-art solutions, *ProfJobs* proved to be a superior alternative that overcame most of the limitations of its counterparts.

The generalizability of the results is limited by the demographic and geographic specificity of the user study. The potential bias of the ground truth dataset for the quality assessment, as well as the limitations associated with the SUS and the UEQ, also pose a threat to the validity. In addition, the SUS, UEQ and the benchmark for data accuracy suggest that there is room for improvement.

The following section outlines possible solutions to these limitations and potential areas of future research.

Future Work.
This thesis establishes a foundation for further exploration and innovation in the field of academic job search. The following paragraphs detail potential areas for future research.

Personalization through NLP. The integration of NLP to offer personalized job recommendations is a remarkable improvement. Future work could focus on the development of advanced NLP algorithms or models that are able to accurately interpret academic profiles, such as skills, publications, and interests to suggest job offers that match their professional profile. This approach could streamline the job search process and provide more efficient and tailored results for the individual user.

Creation of Application Letters using NLP. One promising direction is the use of NLP to support the creation of application letters. The system can create customized cover letters by examining the applicant's publications, skills and the specifics of the job posting. This feature has the potential to significantly reduce the time and effort required to apply for jobs, increasing the chances of landing a position. Future research could investigate the development of a sophisticated text generation model tailored to this goal.

Use of Collected Data for Meta-Analysis. One area with potential for future research is the use of the collected job and vacancy data for a meta-analysis. This analysis can provide insights into the academic job market by examining trends in the number of vacancies, their geographical distribution, the timing, and age demographics of vacancies. The results of this analysis could help job seekers and universities with strategic planning and career development.

Additional Sources for Job Listings. Expanding the sources for job listings would increase the scope of *ProfJobs*. By integrating listings from more job portals and academic networks, the platform could present a more diverse and comprehensive range of opportunities. Future work could focus on partnerships or integrations with additional job boards and academic institutions to aggregate a broader range of job opportunities.

Enhancing User Experience. Future improvements for *ProfJobs* should also focus on usability, as the UEQ suggested, such as introducing a customizable dashboard for tracking applications and receiving personalized job notifications. Implementing a user feedback system could refine a job matching algorithms and provide more relevant suggestions. In addition, the integration of features such as chatbots or virtual consultants could make the user interface more engaging and user-friendly and further enrich the job search experience on the platform.

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Appendix A

A.1 Additional Figures

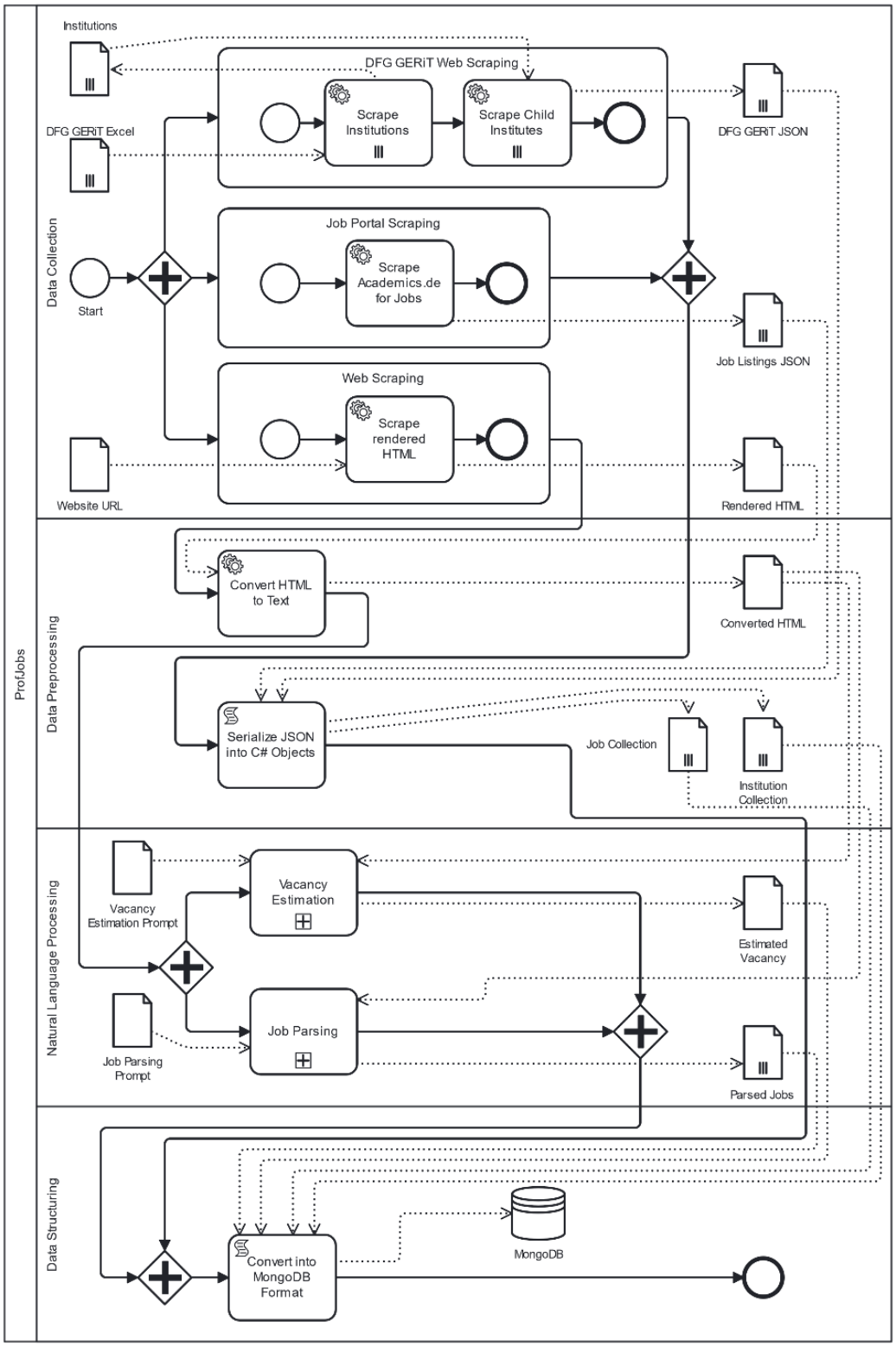


Figure A.1: BPMN of the Implementation

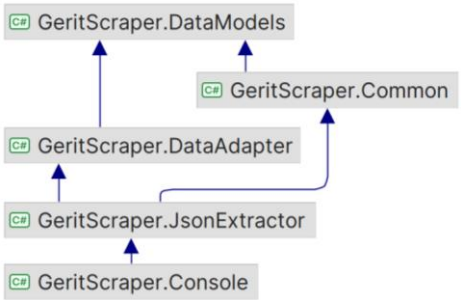


Figure A.2: GERiT Scraper Project Overview

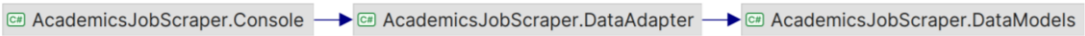


Figure A.3: Academics Scraper Project Overview

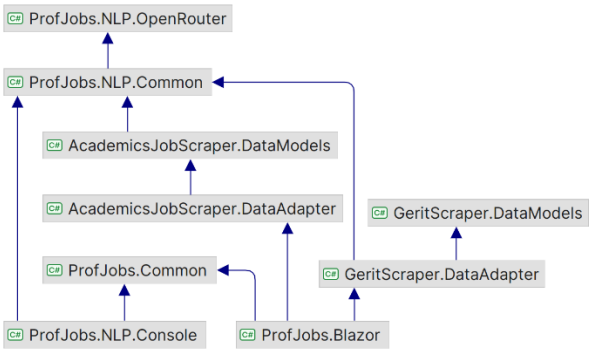


Figure A.4: ProfJobs Project Overview

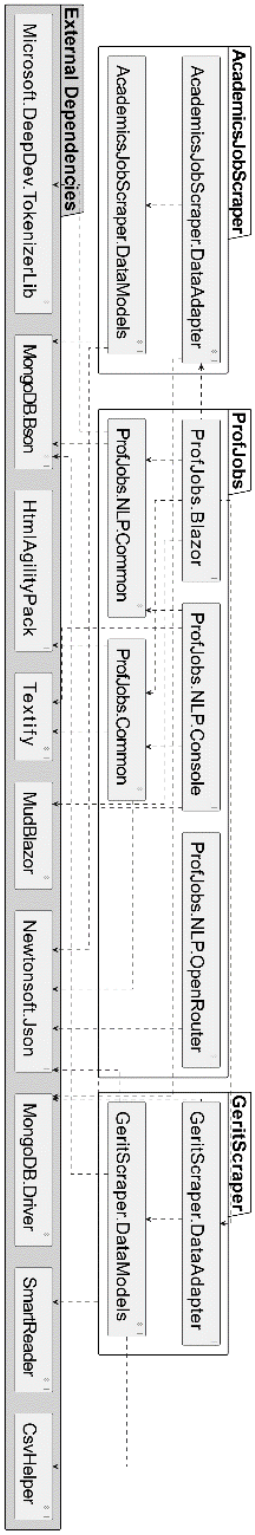
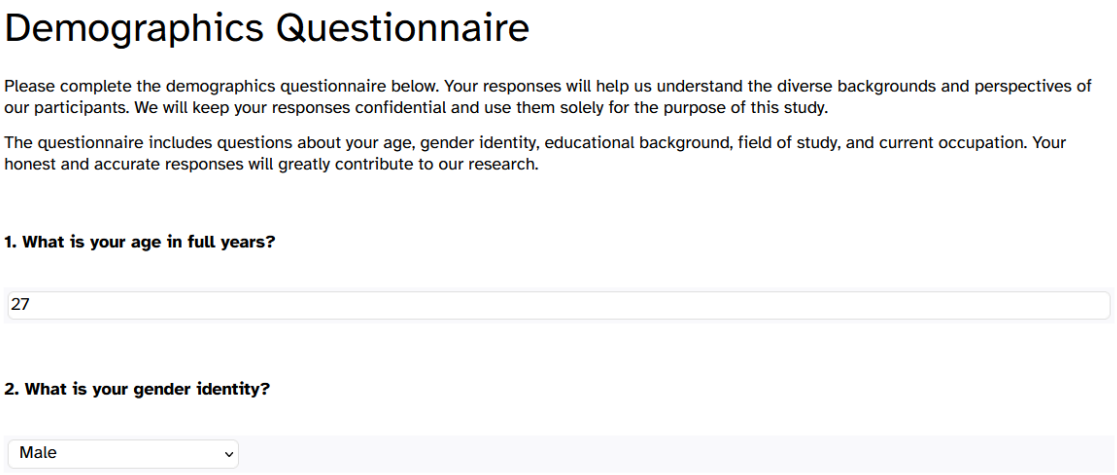


Figure A.5: Project Dependency Diagram of ProfJobs



Demographics Questionnaire

Please complete the demographics questionnaire below. Your responses will help us understand the diverse backgrounds and perspectives of our participants. We will keep your responses confidential and use them solely for the purpose of this study.

The questionnaire includes questions about your age, gender identity, educational background, field of study, and current occupation. Your honest and accurate responses will greatly contribute to our research.

1. What is your age in full years?

27

2. What is your gender identity?

Male

Figure A.6: Part of our Demographics Questionnaire

A.2 Meta Search Engine Scraper Concept

Concept. While search engines are primarily recognized for general search, they can also be used for finding job listings focused on professorships. They are also useful for identifying and enhancing missing data points across our sources. The discovery of current professors' homepages plays a vital role in this context. These websites serve as repositories of essential information that can help in determining the potential availability of a position. Typically, professors display their resumes on their homepages, and occasionally, details such as their birthdate. Analyzing this information can provide valuable insights into the approximate time left before their retirement. Yet, the main challenge is the unstructured format of the data. Thus, further processing is necessary to extract useful information. The intricacies of this extraction process, utilizing natural language processing, will be explored in more depth in a subsequent chapter.

Automated data retrieval from search engines is not a straightforward process. Pre-built APIs offered by search engine companies could be used, however, these application programming interfaces (APIs) have limitations such as costs and restrictions on public access, as highlighted in the chapter 2.2.

To overcome these challenges, our solution requires a custom web scraper designed to retrieve and organize search engine results. This process can be carried out either by utilizing tools that interface directly with search engines through HTTP requests or by utilizing meta search engines. The latter, functioning as search aggregators, query

multiple search engines together, consolidate the received results, and then rank them based on relevance. Platforms such as the open-source meta search engine SearX or its fork, SearXNG [86]. These offer significant advantages, particularly in their ability to withstand rate limits imposed by search providers. Their resilience is attributed to the capacity to combine results from numerous search engines, guaranteeing the display of results even if one or more search engines deny access. Also, these tools excel at managing proxy servers, diversifying the sources of search queries across various IP addresses and user agents.

Another benefit of meta search engines is their ability to display search results in organized layouts, like JSON or XML, with no need for further structuring of the data. These noteworthy characteristics demonstrate why meta search engines are superior to traditional HTTP request-based web scrapers for gathering data from search engines and are therefore good fit for our solution.

Implementation. Our method of data extraction from search engines involved deploying our own instance of SearXNG[30] on a Linux server. This was based on the benefits highlighted during the concept phase (see 3.3), which identified meta search engine usage as the best technique for scraping search engine results. SearXNG, an open-source meta search engine emerged as the ideal candidate because of its aggregation capabilities that bring together search results from multiple sources, while also protecting user privacy.

Scraping search engine data poses a significant challenge due to the rate limits implemented by search engine providers. These limits restrict the amount of traffic and requests that can be sent from a single source within a given time. To circumvent this issue, we programmed our SearXNG instance to rotate among a collection of 100 proxy servers, strategically distributed around the world. This arrangement not only distributes our query load but also conceals our scraping activities, presenting them as genuine search traffic originating from various geographic locations.

Employing SearXNG alongside proxy servers, we successfully collected search engine data without triggering any rate-limiting mechanisms. This technique guaranteed a consistent stream of data necessary for our platform, while maintaining the responsiveness and reliability of the queried search engines.

[30] https://github.com/searxng/searxng

A.3 NLP Prompts

```
Role: System
Analyze the document that will be provided in the next message (CV or
publication list) to estimate the age of the individual. Follow these
steps:
1. Extract relevant dates from the document, focusing on:
   - Year of graduation (high school, undergraduate, graduate).
   - Dates of employment.
   - Publication dates, if a publication list is provided.
2. Estimate the age:
    - Assume high school graduation at 18 - 19 years old, undergraduate
    degree completion at around 22, and graduate degree completion at
    around 25-27. Publication is uncommon when not holding an
    undergraduate degree.
   - Consider the earliest and latest dates in the professional experience
and publications.
   - Use the current year for calculation.
3. Determine the success status:
    - If enough data is available to make a reasonable estimate, set
    success to true.
    - If data is insufficient or contradictory, set success to false and
    provide a reason in the status.
4. Output a JSON object with the following structure:
   {
     "success": [true/false],
     "status": "[explanation if success is false, else 'Age    estimated
    successfully']",
     "age": [estimated age as integer if successful, else null]
   }. ONLY OUTPUT THIS JSON OBJECT, NOTHING ELSE!
```

Listing A.1: Vacancy Estimation NLP Prompt

```
Role: System
I am about to send you a text containing multiple job listings for
professorships. Your task is to analyze this text and extract specific
information from each job listing. The output must be formatted as an
array of JSON objects, with each object representing a single job listing.
It is crucial that the output is a JSON array, not just a single JSON
object or a text message. Here is the structure each JSON object in the
array should follow:
{
  "JobListings": [
    {
      "Country": "value or null",
      "State": "value or null",
      "University": "value or null",
      "TypeOfInstitution": "value or null",
      "Faculty": "value or null",
      "Institute": "value or null",
      "Chair": "value or null",
      "Professorship": "value or null",
      "URLs": "value or null",
      "JobHolder": "value or null",
      "AppointmentDate": "value (datetime) or null",
      "Grouping": "value (W1,W2,W3 or TV-L) or null",
      "VacancyStatus": "value or null",
      "VacancyDate": "value (datetime) or null",
      "ApplicationDeadline": "value (datetime) or null"
    },
    // Additional job listings in the same format
  ]
}
Each JSON object represents a different job listing. If a piece of
information is not available for a listing, use null for that field.
Please ensure that the entire output is a JSON array containing these
objects. This format is non-negotiable and must be strictly adhered to. I
will now send you the text with the job listings.
```

Listing A.2: Custom Job Parsing NLP Prompt

A.4 Demographics Questionnaire

1. **What is your age?**

2. **What is your gender identity?**
    a. Male
    b. Female
    c. Non-binary/Third gender
    d. Prefer not to say
    e. Other

3. **What is your field of study or academic background?**
    a. Arts & Humanities (e.g., Literature, Philosophy, History)
    b. Social Sciences (e.g., Psychology, Sociology, Anthropology)
    c. Natural Sciences (e.g., Biology, Chemistry, Physics)
    d. Formal Sciences (e.g., Mathematics, Logic)
    e. Computer Science
    f. Applied Sciences (e.g., Engineering, Environmental Science, Agriculture)
    g. Medical & Health Sciences (e.g., Medicine, Nursing, Pharmacy)
    h. Business & Management (e.g., Finance, Marketing, Human Resources)
    i. Education (e.g., Early Childhood Education, Educational Psychology)
    j. Law (e.g., Criminal Law, International Law, Corporate Law)
    k. Others (for fields not listed above)
    l. None (for participants without an academic background)

4. **What is your current occupation?**
    a. PhD Student
    b. Postdoc / PhD
    c. Assistant Professor
    d. Associate Professor
    e. Professor
    f. Other

5. **What is your mother tongue?**
    a. German
    b. Non-German

6. **Have you used a translation tool?**
    a. Yes
    b. No

7. **In which country are you currently based?**
    - List of 98 common countries & the item Other

A.5 Efficiency Comparison Study

The purpose of this study is to compare the time efficiency of using the ProfJobs platform versus manual methods for compiling job listings. The independent variable is the method used for compiling job listings, which includes two categories: the use of the *ProfJobs* platform and the manual searching method. The study aims to determine which method is more time efficient. The dependent variable is the time efficiency, measured as the search time to compile a list of job listings. Efficiency is a crucial metric that determines the practical utility of the application. A more efficient process for creating a list of job listings not only saves time but also positions *ProfJobs* as a superior alternative to traditional job search methods.

$\mathbf{H}_{\mathbf{eff}}$ Utilizing the *ProfJobs* platform for compiling job listings does significantly reduce the search time compared to manual searching.

Material. The focus of this study is *ProfJobs*, a web application prototype that gathers and facilitates the exploration of job listings for professorships which was implemented in this thesis (see 4). A scenario (see Listing A.3) is presented to participants where they are positioned as a PhD in computer science from Germany who seeks a W1 or W2 professorship in the field of web engineering. This scenario is essential to guide the participants' search process and maintain consistency in their tasks. A detailed description of the tasks and scenario, along with a page for recording time and submitting feedback, was provided using the SoSci platform.

Procedure. The study's procedure comprises two primary tasks that aim to assess the effectiveness of various job search methods. To minimize any potential biases related to task order, the participants' engagement with *ProfJobs* and the manual method is randomized.

- **Finding Positions using ProfJobs:** Participants must utilize *ProfJobs* to find and export five job positions that meet the criteria outlined in the scenario.
- **Finding Positions using Traditional Methods:** The task was then replicated by the participants using conventional job search techniques, such as basic internet searches, job search engines, and university career portals.

Participants complete the task using both methods, and the search time for each method will be tracked by the participants. After each task the participants provide time taken through a custom questionnaire.

Instrumentation. The instrumentation for this study includes time tracking and a custom questionnaire. Participants are instructed to meticulously record the time taken to

complete each task using a stopwatch or a similar timing device. Following each task, this time data is submitted through a custom questionnaire on the SoSci[31] platform.

```
Background: You are a PhD in Computer Science from a German university.
You finished your doctoral studies and now your aim is to find a job in
academia. You are seeking a professorship that fits your passions and
expertise.

Your Objective: As someone who is passionate about web-related fields, you
are searching for job opportunities that align with your academic and
research interests. Your objective is to identify potential job openings
for W1 or W2 professorships in the field of computer science, with a
particular emphasis on areas relevant to web engineering.

Task Details: Your objective is to create a table (e.g. Excel sheet) of
five (5) potential job positions that you would like to pursue. Track the
time it takes you to create the table (e.g. using a stopwatch, your phone,
or simply the clock). Ensure that the table contains the following vital
details for each job position:
   •University Name: Where is the position located?
   •Faculty: Which department or faculty is offering the position?
   •Job Title: What is the official title of the position?
   •Application Link or Email: How and where can you apply?
   •Application Deadline: When is the last date you can apply?
   •Position Description: What are the key details about the job?
```

Listing A.3: Efficiency Comparison Scenario

A.6 System Usability Scale

1) I think that I would like to use this system frequently.
2) I found the system unnecessarily complex.
3) I thought the system was easy to use.
4) I think that I would need the support of a technical person to be able to use this system.
5) I found the various functions in this system were well integrated.
6) I thought there was too much inconsistency in this system.
7) I would imagine that most people would learn to use this system very quickly.
8) I found the system very cumbersome to use.

[31] https://www.soscisurvey.de/

9) I felt very confident using the system.

10) I needed to learn a lot of things before I could get going with this system.

A.7 UEQ Results

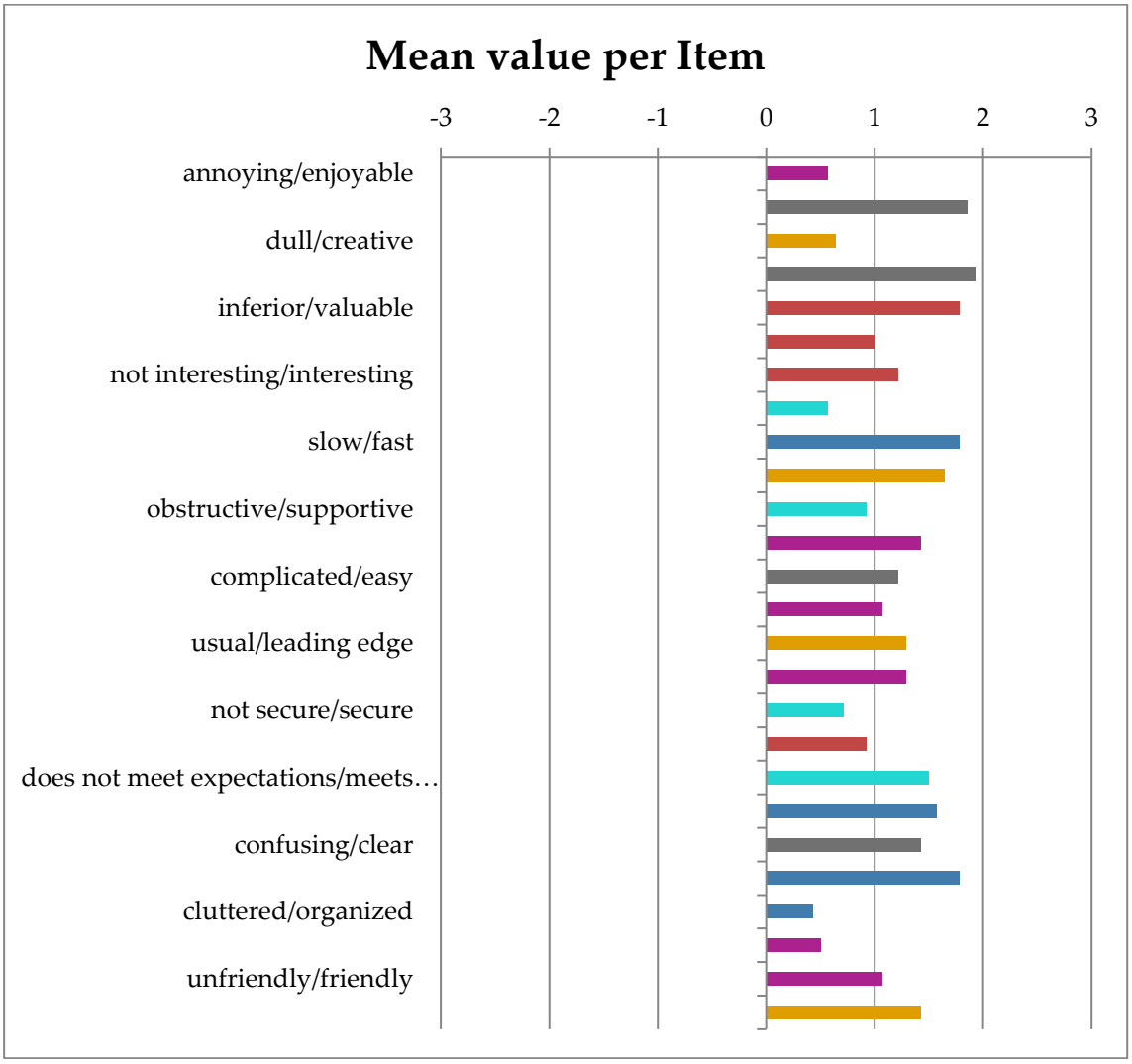


Figure A.7: UEQ: Mean Values per Item

Confidence intervals (p=0.05) per scale						
Scale	Mean	Std. Dev.	N	Confidence	Confidence interval	
Attractiveness	0.988	1.644	14	0.861	0.127	1.849
Perspicuity	1.607	1.580	14	0.828	0.779	2.435
Efficiency	1.393	1.055	14	0.553	0.840	1.945
Dependability	0.929	1.592	14	0.834	0.095	1.762
Stimulation	1.232	1.353	14	0.709	0.523	1.941
Novelty	1.250	1.205	14	0.631	0.619	1.881

Table A.1: UEQ: Confidence Intervals and Standard Deviation

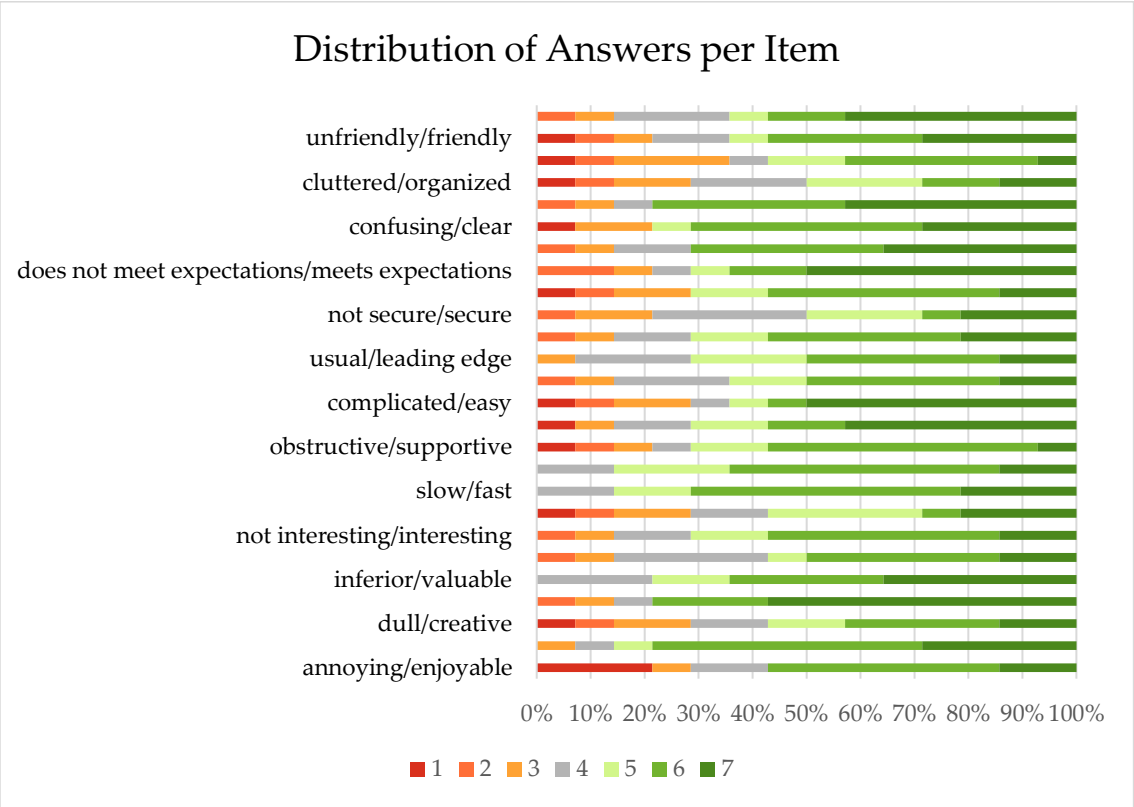


Figure A.8: UEQ: Distribution of Answers per Item

A.8 Data Quality Assessment

University	Number of Job Listings
Ludwig-Maximilians-Universität München (LMU)	5
Baden-Wuerttemberg Cooperative State University (DHBW)	12
Universität Hamburg	5
University of Cologne	2
University of Münster	4
Friedrich-Alexander-Universität Erlangen-Nürnberg	2
University of Bonn	2
Goethe University Frankfurt	3
Chemnitz University of Technology (CUT)	1

Table A.2: Overview of Universities in Data Quality Assessment

Id	Deadline	Apply Email	Employer	Faculty	Apply URL	Title	Salary Group	Average Distance	Median Distance
4	0.00	0.00	6.00	30.00	0.00	0.00	0.00	19.38	0.00
5	0.00	0.00	6.00	30.00	0.00	0.00	0.00	24.00	0.00
7	2.00	0.00	38.00	19.00	71.00	31.00	1.00	33.00	25.00
8	2.00	0.00	46.00	19.00	70.00	0.00	0.00	30.63	10.50
9	2.00	0.00	43.00	19.00	99.00	51.00	0.00	34.63	31.00
10	10.00	0.00	52.00	16.00	72.00	0.00	0.00	35.00	13.00
12	0.00	0.00	38.00	19.00	71.00	8.00	0.00	29.50	13.50
15	10.00	0.00	48.00	16.00	22.00	0.00	0.00	24.25	13.00
16	5.00	0.00	46.00	19.00	42.00	11.00	0.00	29.75	15.00
17	0.00	0.00	45.00	19.00	8.00	0.00	0.00	22.25	4.00
18	10.00	26.00	0.00	0.00	0.00	48.00	1.00	19.88	5.50
19	10.00	26.00	0.00	0.00	0.00	42.00	0.00	17.38	5.00
20	0.00	0.00	0.00	31.00	71.00	0.00	1.00	25.88	0.50
23	1.00	0.00	0.00	34.00	11.00	0.00	12.00	12.25	6.00
24	2.00	0.00	0.00	34.00	11.00	0.00	1.00	11.00	1.50
25	10.00	30.00	4.00	33.00	0.00	1.00	0.00	13.25	7.00
26	10.00	24.00	4.00	22.00	0.00	87.00	0.00	21.88	16.00
27	10.00	21.00	4.00	29.00	0.00	95.00	0.00	23.38	15.50
28	10.00	21.00	4.00	21.00	0.00	67.00	0.00	18.88	15.50
29	1.00	18.00	0.00	0.00	0.00	0.00	0.00	3.75	0.00
30	1.00	16.00	0.00	0.00	0.00	0.00	3.00	3.88	0.50
31	2.00	56.00	34.00	19.00	0.00	0.00	0.00	26.50	10.50
32	10.00	56.00	34.00	28.00	0.00	0.00	0.00	28.63	19.00
33	10.00	43.00	0.00	11.00	0.00	6.00	0.00	13.63	8.00
34	10.00	33.00	0.00	11.00	0.00	5.00	0.00	12.38	7.50
35	10.00	32.00	0.00	11.00	0.00	19.00	0.00	13.75	10.50
36	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
							Overall	**20.32**	**8.00**

Figure A.9: Data Quality: Distance Matrix

Id	Deadline	Apply Email	Employer	Faculty	Apply URL	Title	Salary Group	Average Similarity	Median Similarity
4	100.00	100.00	100.00	0.00	100.00	86.36	100.00	83.77	100.00
5	100.00	100.00	100.00	0.00	100.00	86.36	100.00	83.77	100.00
7	59.21	28.28	80.00	0.00	100.00	29.63	50.00	49.59	50.00
8	100.00	29.29	80.00	0.00	100.00	14.81	100.00	60.59	80.00
9	27.14	0.00	80.00	0.00	100.00	20.37	100.00	46.79	27.14
10	100.00	27.27	0.00	0.00	100.00	21.21	100.00	49.78	27.27
12	84.91	28.28	100.00	0.00	100.00	29.63	100.00	63.26	84.91
15	100.00	77.78	0.00	0.00	100.00	11.11	100.00	55.56	77.78
16	80.70	58.00	50.00	0.00	100.00	14.81	100.00	57.65	58.00
17	100.00	92.00	100.00	0.00	100.00	16.67	100.00	72.67	100.00
18	26.15	100.00	0.00	100.00	0.00	100.00	50.00	53.74	50.00
19	20.75	100.00	0.00	100.00	0.00	100.00	100.00	60.11	100.00
20	100.00	0.00	100.00	0.00	100.00	100.00	50.00	64.29	100.00
23	100.00	81.36	90.00	20.93	100.00	100.00	14.29	72.37	90.00
24	100.00	81.36	80.00	20.93	100.00	100.00	66.67	78.42	81.36
25	98.31	100.00	0.00	0.00	0.00	80.95	100.00	54.18	80.95
26	39.58	100.00	0.00	0.00	0.00	80.95	100.00	45.79	39.58
27	45.71	100.00	0.00	0.00	0.00	80.95	100.00	46.67	45.71
28	27.17	100.00	0.00	0.00	0.00	80.95	100.00	44.02	27.17
29	100.00	100.00	90.00	100.00	0.00	100.00	100.00	84.29	100.00
30	100.00	100.00	90.00	100.00	0.00	100.00	40.00	75.71	100.00
31	100.00	100.00	80.00	0.00	0.00	26.09	100.00	58.01	80.00
32	100.00	100.00	0.00	0.00	0.00	26.09	100.00	46.58	26.09
33	92.94	100.00	0.00	64.52	0.00	100.00	100.00	65.35	92.94
34	89.80	100.00	0.00	74.42	0.00	100.00	100.00	66.32	89.80
35	74.67	100.00	0.00	63.33	0.00	100.00	100.00	62.57	74.67
36	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
							Overall	63.03	80.95

Figure A.10: Data Quality: Similarity Matrix

A.9 Usability Evaluation Scenarios

Scenario: You are a postdoc in the field of software engineering. You have been working full-time at a German university in the field of computer science at a professorship for software engineering since over 3 years. Your plan is to have your own professorship in software engineering at a German university in the next few years.

Now imagine it's Wednesday evening, you've done all your work for the day, you're relaxed and you want to use a platform specifically tailored for the search of professorships for postdocs that you've recently found to look for potential professorship positions.

Task: Your task is to use the ProfJobs platform to identify three (3) vacant positions in software engineering. The platform is available at https://profjobs.codelu.eu.

1) Find and mark three (3) professorship job positions in software engineering as favorites.

2) Export all of your favorite jobs as Excel sheet.

3) Share your most favorite job position below.

4) Upload the exported Excel sheet with all of your favorite job positions.

Note: There are no wrong answers. We do not assess your performance, skills or knowledge.

Enter the following information of your most favorite job position:

Job Title
University Name
Salary Group

Listing A.4: Usability Evaluation Scenario 1 (Job Search)

Scenario: While having lunch together in the canteen, your colleague told you that he believes a professor of software engineering is close to reaching the retirement age and that he expects him to retire within the next few years. As you are looking for a suitable professorship in software engineering, you are interested in his retirement for applying early to this position.

Since you both don't know his exact age and therefore the year he will retire, you want to use ProfJobs' vacancy estimator feature to estimate his age based on his online resume.

Task: Go to the ProfJobs vacancy estimation page located at https://profjobs.codelu.eu/vacancies to estimate the professor's age and thus also his retirement. Insert the following URL to his online resume http://www1.inf.tu-dresden.de/~ua1/ for estimating the age using ProfJobs.

After estimating his age using ProfJobs, submit the results below.

Note: There are no wrong answers. We do not assess your performance, skills or knowledge.

Enter the results of the vacancy estimation of the professor:

Full Name
Predicted Vacancy Year
Age

Listing A.5: Usability Evaluation Scenario 2 (Vacancy Estimation)

Appendix B

The print version of this thesis comes with an SD card containing additional materials.

B.1 Thesis

- Contains the thesis in PDF.

B.2 Experiments

B.2.1 Survey project

- Contains the survey project, which can be imported to SoSci.

B.2.2 User Study

- Contains the raw data as well as the data analytics tools used.

B.2.3 Quality Assessment

- Contains the raw data as well as the data analytics tools used.

B.3 Source code

B.3.1 ProfJobs

- Contains the source code of ProfJobs.

B.3.2 Academics Scraper

- Contains the source code of the academics.com Scraper.

B.3.3 Web Scraper API

- Contains the source code of the Web Scraper API.

B.3.4 DFG GERiT Scraper

- Contains the source code of the DFG GERiT Scraper.

B.3.5 Search Engine API

- Contains the source code for the Search Engine API.

Acknowledgment of Language and Writing Tools Utilized

Throughout the creation of this thesis, DeepL Translate[32] was used to facilitate the translation of text and words from German to English and to ensure accuracy and coherence when conveying information in different languages. In addition, DeepL Write[33] was used to improve grammatical integrity and refine the tone of the text to create a more reader-friendly document.

[32] https://www.deepl.com/translator
[33] https://www.deepl.com/write

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